



*Mechanical
Engineering
Department*

**KONYA
TÜRKİYE**



ENGINEERING MECHANICS **STATICS**

| SIXTH EDITION |

J. L. MERIAM

L. G. KRAIGE



*Mechanical
Engineering
Department*

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EQUILIBRIUM

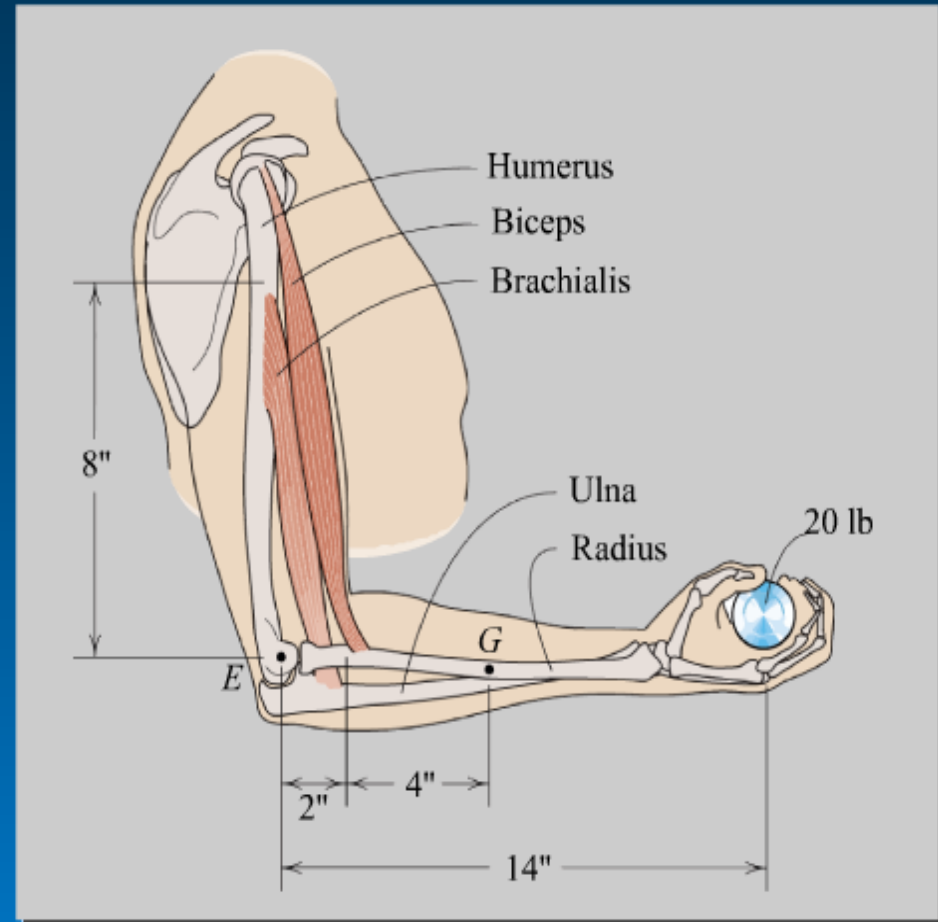
3. Equilibrium

Introduction

Free-Body Diagrams

Equilibrium in Two Dimensions

Equilibrium in Three Dimensions



Introduction

A body is said to be in *equilibrium* when the external force-couple system acting on it is zero. This means that both the resultant force $\mathbf{R} = \Sigma \mathbf{F}$ and the couple $\Sigma \mathbf{M} = \Sigma \mathbf{r} \times \mathbf{F}$ are zero. Thus, the conditions for equilibrium are

$$\Sigma \mathbf{F} = 0 \quad (\text{ensures translational equilibrium})$$

$$\Sigma \mathbf{M} = 0 \quad (\text{ensures rotational equilibrium})$$

The scalar equivalents of the above vector equations (in rectangular coordinates) are:

$$\Sigma F_x = 0 \quad (1) \qquad \Sigma M_x = 0 \quad (4)$$

$$\Sigma F_y = 0 \quad (2) \qquad \Sigma M_y = 0 \quad (5)$$

$$\Sigma F_z = 0 \quad (3) \qquad \Sigma M_z = 0 \quad (6)$$

For equilibrium of a rigid-body in two dimensions (in the x - y plane), only Eqs. (1), (2), and (6) are normally used. The moment equations can be referenced to axes through any point on or off the body.



Free-Body Diagram

A free-body diagram (FBD) is *essential* in the solution of all equilibrium and kinetics problems. The elements of a good FBD are:

- (1) The body (or systems of bodies) must be identified, isolated from its environment, and sketched. Avoid excessive detail, especially *internal* detail. Generally, an *outline* of the body is all that is necessary. Use a straightedge/circle template.

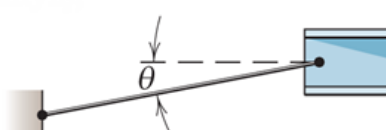
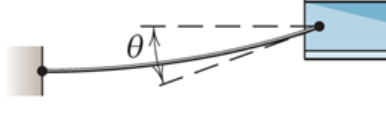
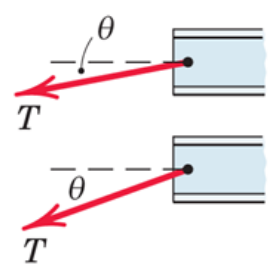
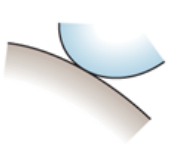
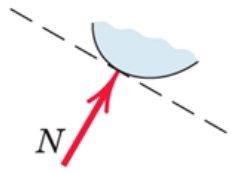
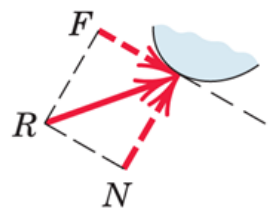
2. All *external* forces and moments acting on the body should be clearly shown on the sketch. If a force or moment is known, use its value as the label. If a force or moment is unknown, give it a variable as a label.
3. Label pertinent points and dimensions. Include any items needed for writing the equilibrium equations.
4. Indicate a coordinate system. Choose a coordinate system that will be convenient when writing the equilibrium equations.



Let us expand on item 2 above. In replacing a mechanical support by forces and moments, the following suggestions are offered:

- a. Imagine small linear displacements of the body at the point of support. If there is resistance to movement in a given direction, then a reaction force in that direction is necessary. Repeat for all three coordinate directions.
- b. Similarly, for each support, imagine rotational displacements about all three axes. If there is resistance to a given rotation, a moment about that axis is necessary.

Apply the above thinking to the examples of the figure below.

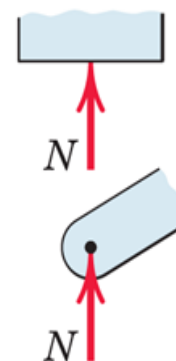
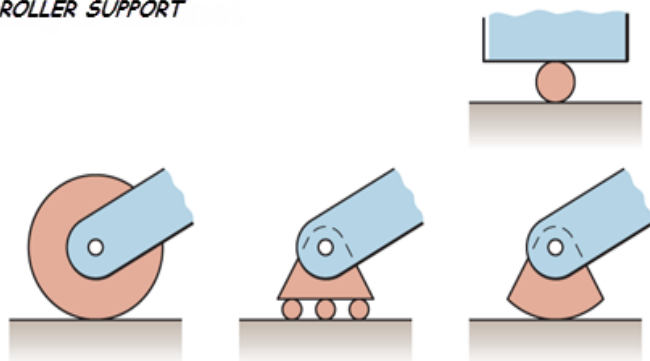
MODELLING THE ACTION OF FORCES IN TWO DIMENSIONAL ANALYSIS	
TYPE CONTACT AND FORCE ORIGIN	ACTION ON BODY TO BE ISOLATED
1. FLEXIBLE CABLE, BELT, CHAIN OR ROPE WEIGHT OF CABLE NEGLECTIBLE  WEIGHT OF CABLE NOT NEGLECTIBLE 	 FORCE EXERTED BY A FLEXIBLE CABLE IS ALWAYS A TENSION AND AWAY FROM THE BODY IN THE DIRECTION OF THE CABLE
2. SMOOTH SURFACES 	 CONTACT FORCE IS COMPRESSIVE AND IS NORMAL TO THE SURFACE
3. ROUGH SURFACES 	 ROUGH SURFACES ARE CAPABLE OF SUPPORTING A TANGENTIAL COMPONENT F (FRICTIONAL FORCE) AS WELL AS A NORMAL COMPONENT N OF THE RESULTANT CONTACT FORCE R

MODELLING THE ACTION OF FORCES IN TWO DIMENSIONAL ANALYSIS

TYPE CONTACT AND FORCE ORIGIN

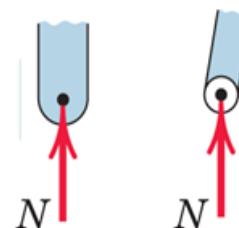
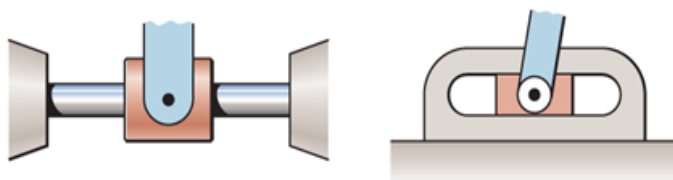
ACTION ON BODY TO BE ISOLATED

4. ROLLER SUPPORT



ROLLER, ROCKER OR BALL
SUPPORTS TRANSMITS A
COMPRESSIVE FORCE NORMAL
TO THE SUPPORTING SURFACE

5. FREELY SLIDING GUIDE



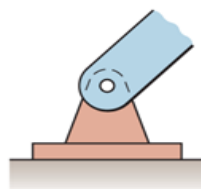
COLLAR OR SLIDER FREE TO MOVE
ALONG SMOOTH GUIDES; CAN
SUPPORT FORCE NORMAL TO
GUIDE ONLY

MODELLING THE ACTION OF FORCES IN TWO DIMENSIONAL ANALYSIS

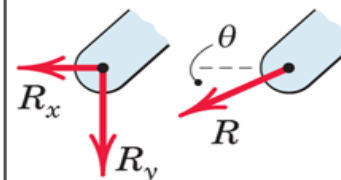
TYPE CONTACT AND FORCE ORIGIN

ACTION ON BODY TO BE ISOLATED

6. PIN CONNECTION

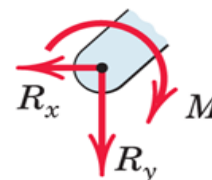


PIN FREE TO TURN



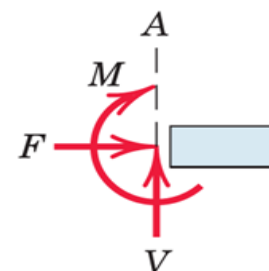
A FREELY HINGED PIN CONNECTION IS CAPABLE OF SUPPORTING A FORCE IN ANY DIRECTION IN THE PLANE NORMAL TO THE AXIS; USUALLY SHOWN AS R_x AND R_y .

PIN NOT FREE TO TURN



A PIN NOT FREE TO TURN MAY ALSO SUPPORT A COUPLE M .

7. BUILD-IN OR FIXED SUPPORT



A BUILD-IN OR FIXED SUPPORT IS CAPABLE OF SUPPORTING AN AXIAL FORCE F , A TRANSVERSE FORCE V (SHEAR FORCE) AND A COUPLE M (BENDING MOMENT) TO PREVENT ROTATION

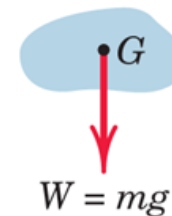
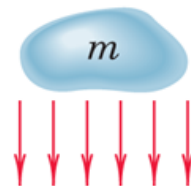


MODELLING THE ACTION OF FORCES IN TWO DIMENSIONAL ANALYSIS

TYPE CONTACT AND FORCE ORIGIN

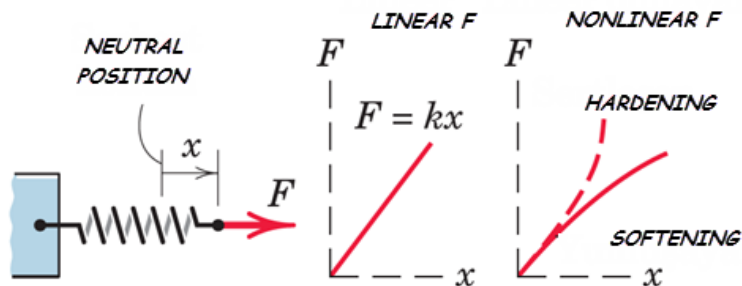
ACTION ON BODY TO BE ISOLATED

8. GRAVITATIONAL ATTRACTION



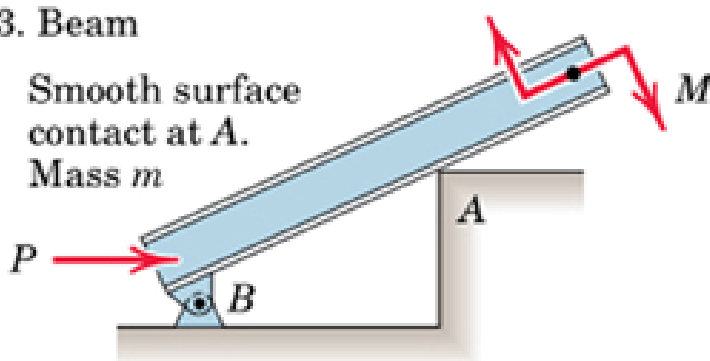
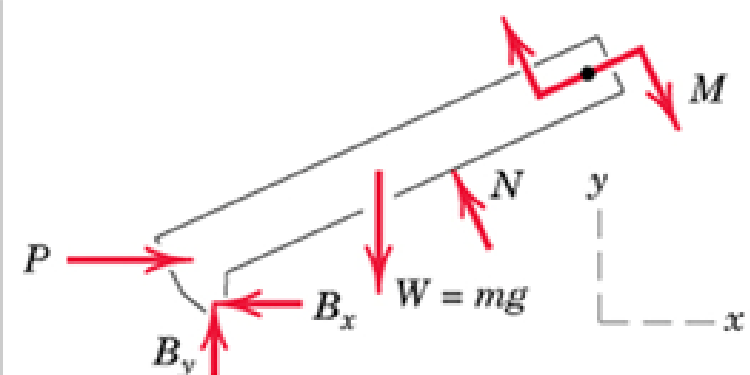
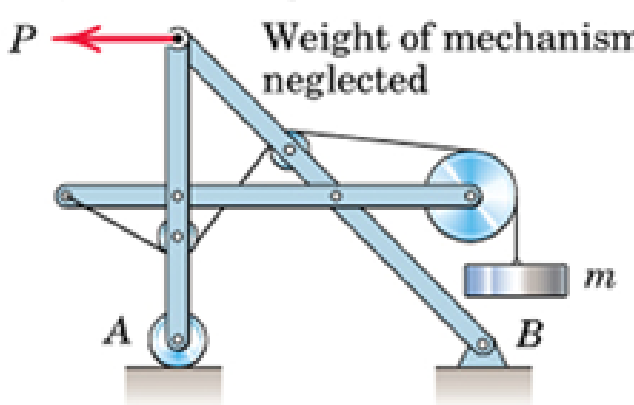
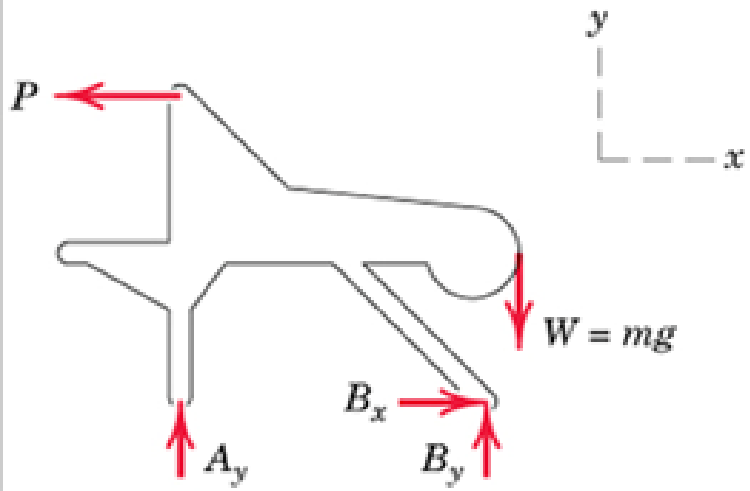
THE RESULTANT OF GRAVITATIONAL ATTRACTION ON ALL ELEMENTS OF A BODY OF MASS m IS THE WEIGHT $W=mg$ AND ACTS TOWARD THE CENTER OF THE EARTH THROUGH THE CENTER MASS G

9. SPRING ACTION



SPRING FORCE IS TENSILE IF SPRING IS STRETCHED AND COMPRESSIVE IF COMPRESSED.
FOR A LINEARLY ELASTIC SPRING THE STIFFNESS k IS THE FORCE REQUIRED TO DEFORM THE SPRING A UNIT DISTANCE

SAMPLE FREE-BODY DIAGRAMS

Mechanical System	Free-Body Diagram of Isolated Body
3. Beam Smooth surface contact at A. Mass m 	
4. Rigid system of interconnected bodies analyzed as a single unit Weight of mechanism neglected 	

Equilibrium in Two Dimensions

Solution Outline For 2-D Equilibrium Problems

1. Choose body (or system of bodies) whose equilibrium is to be studied. Construct its FBD.
2. Apply the governing equations. For 2-D problems,

$$\Sigma F_x = 0 \quad :$$
 (1)

$$\Sigma F_y = 0 \quad :$$
 (2)

$$\Sigma M_o = 0 \quad :$$
 (3)

Write the governing equations, then, to the right of colon, apply it specifically to the FBD.

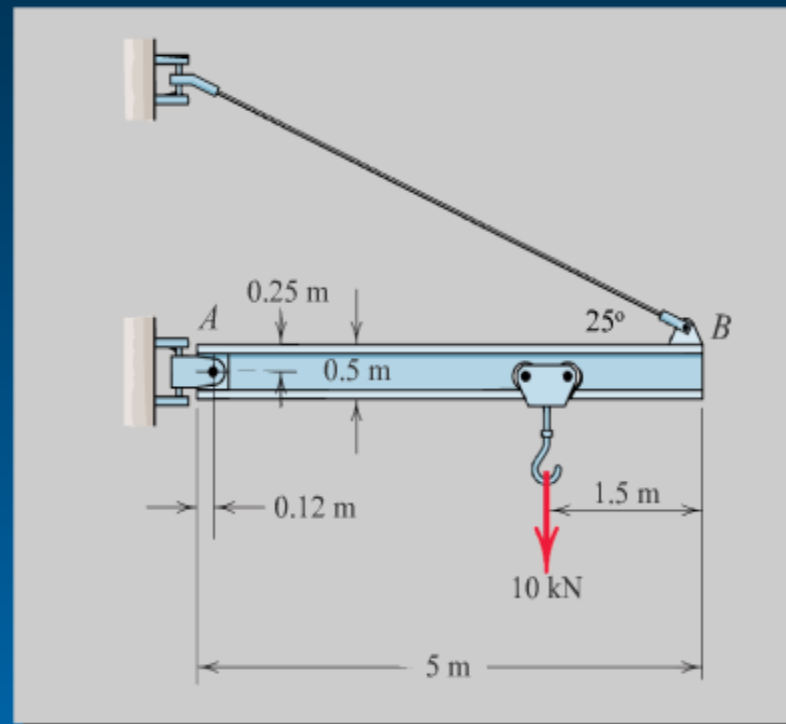


3. Count and list the unknowns in your equations. For n unknowns in m governing equations, if
- $n \leq m$: OK. Solve for the unknowns.
- $n > m$: You must write additional relationships (such as $F = \mu N$), bringing the number of equations equal to the number of unknowns, or else conclude that the body is statically indeterminate.
4. Apply your common sense. Are the magnitudes and directions of your results reasonable? If the problem is one of a variable, rather than numeric, nature, check the dimensions (units) of your results.



Example 1

Determine the magnitude T of the tension in the supporting cable and the magnitude of the force on the pin at A for the jib crane subjected to the 10 kN load shown. The weight of the uniform beam is 6 kN.

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Click on "Next" to show the FBD



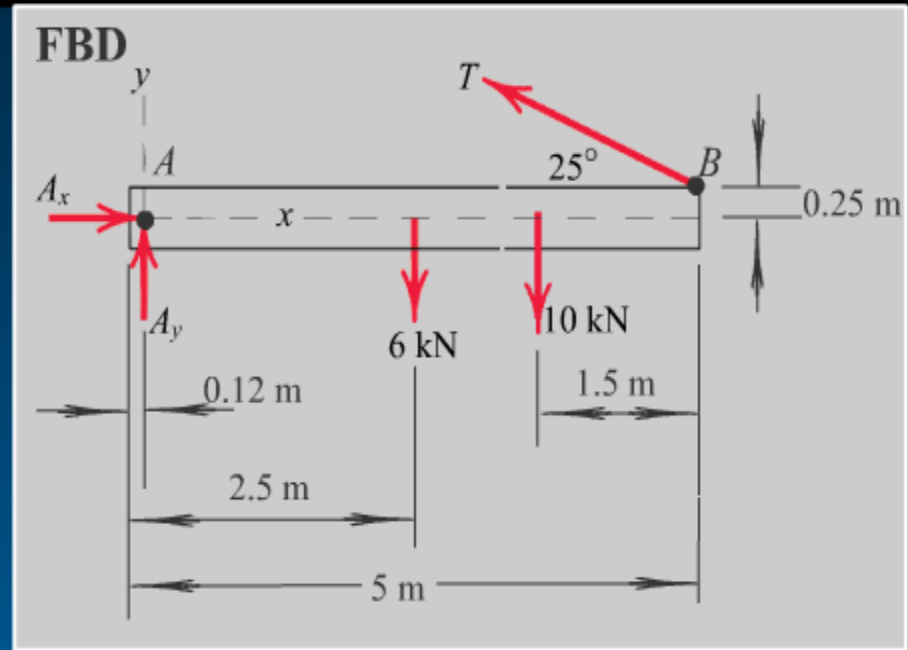
3/4



$$\Sigma F_x = 0 :$$

$$\Sigma F_y = 0 :$$

$$\Sigma M_A = 0 : \quad + \curvearrowright$$



[Back](#)

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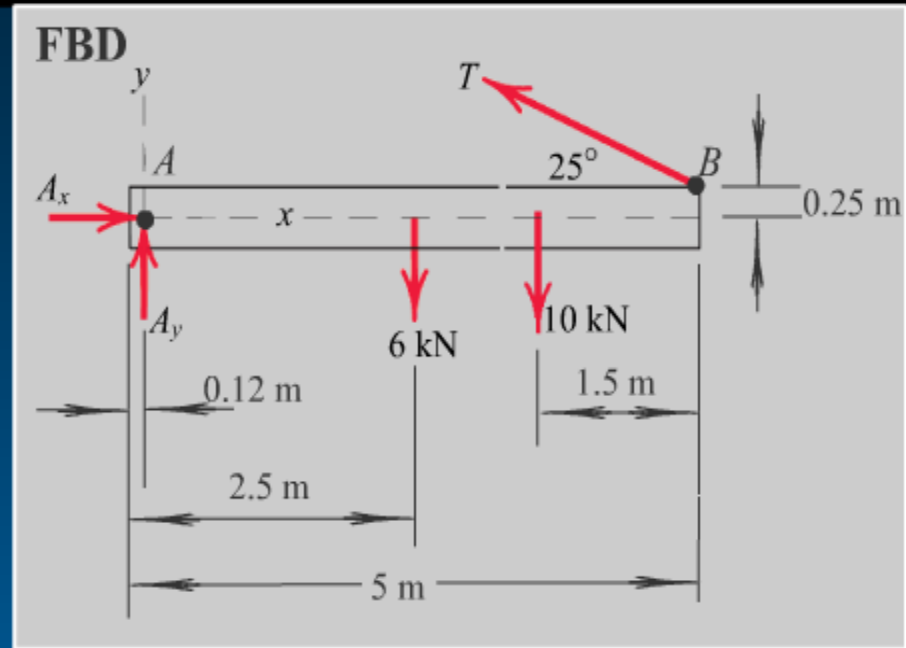


$$\Sigma F_x = 0 :$$

$$A_x - T \cos 25^\circ = 0$$

$$\Sigma F_y = 0 :$$

$$\Sigma M_A = 0 : +\curvearrowright$$



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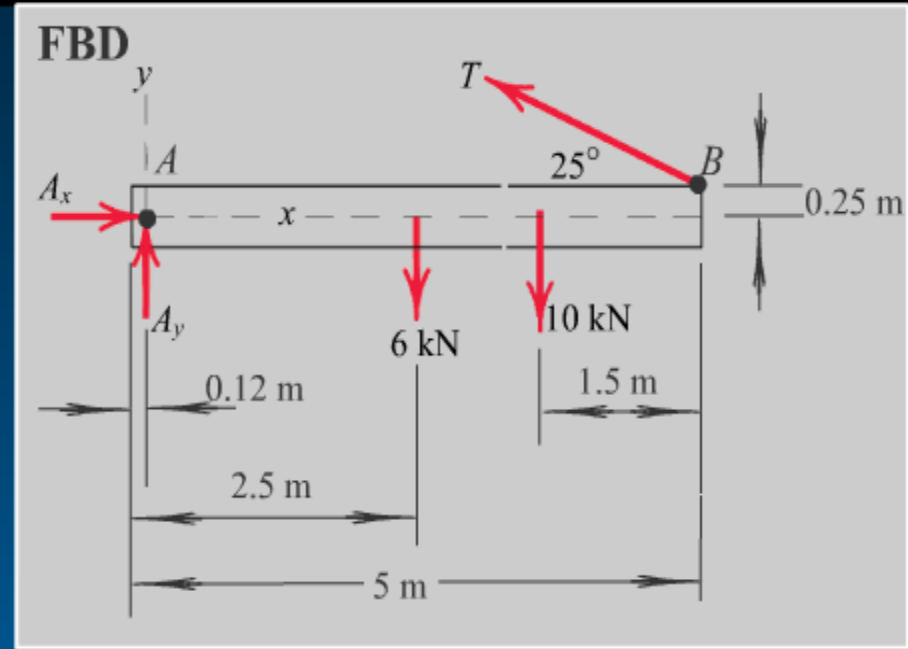
$$\Sigma F_x = 0 :$$

$$A_x - T \cos 25^\circ = 0$$

$$\Sigma F_y = 0 :$$

$$A_y + T \sin 25^\circ - 6 - 10 = 0$$

$$\Sigma M_A = 0 : +\curvearrowright$$



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$$\Sigma F_x = 0 :$$

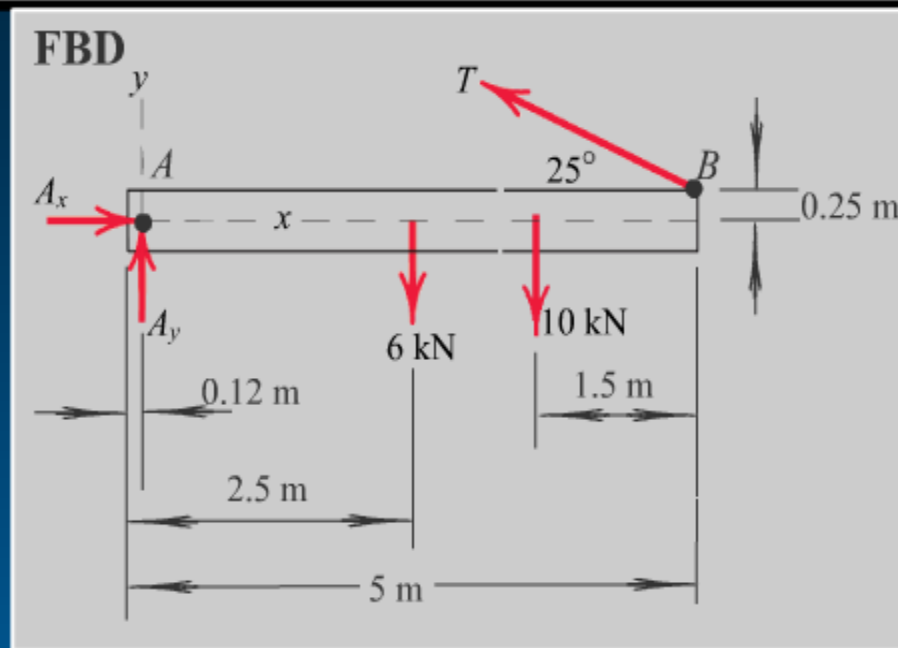
$$A_x - T \cos 25^\circ = 0$$

$$\Sigma F_y = 0 :$$

$$A_y + T \sin 25^\circ - 6 - 10 = 0$$

$$\Sigma M_A = 0 : +\curvearrowright$$

$$(T \cos 25^\circ) 0.25 + (T \sin 25^\circ) (5 - 0.12) - 10 (5 - 1.5 - 0.12) - 6 (2.5 - 0.12) = 0$$



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Click on "Back"/"Next" to show the governing equations



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$$\Sigma F_x = 0 :$$

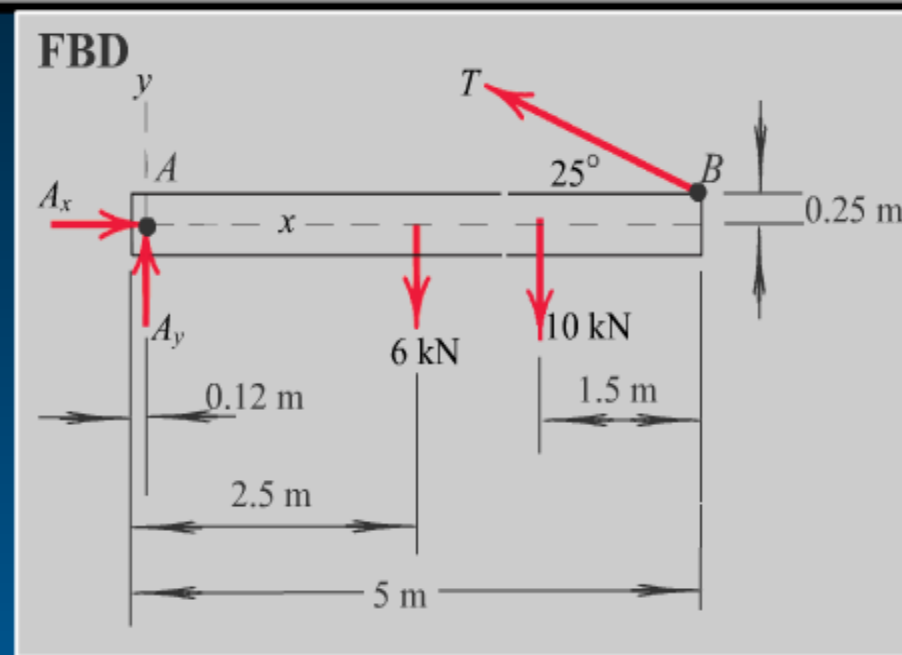
$$A_x - T \cos 25^\circ = 0$$

$$\Sigma F_y = 0 :$$

$$A_y + T \sin 25^\circ - 6 - 10 = 0$$

$$\Sigma M_A = 0 : +\curvearrowright$$

$$(T \cos 25^\circ) 0.25 + (T \sin 25^\circ)(5 - 0.12) - 10(5 - 1.5 - 0.12) - 6(2.5 - 0.12) = 0$$



Solution:

$$A_x = 19.04 \text{ kN} \quad A_y = 7.13 \text{ kN} \quad T = 21.0 \text{ kN}$$

$$A = \sqrt{A_x^2 + A_y^2} = \sqrt{(19.04)^2 + (7.13)^2} = 20.3 \text{ kN}$$

Back

Click "Back" to dismiss solution



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Equilibrium in Three Dimensions

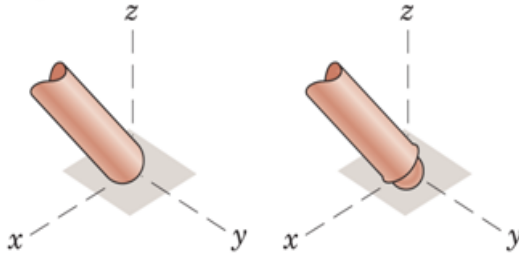
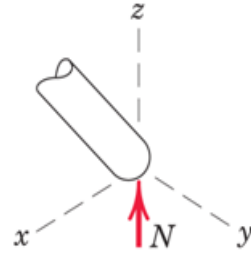
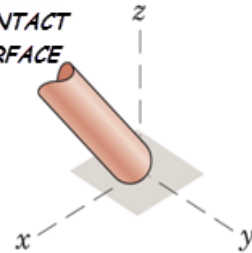
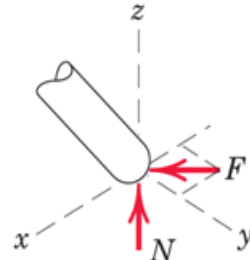
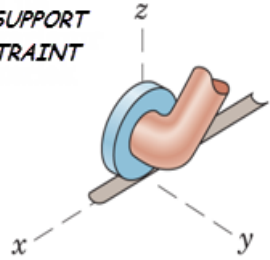
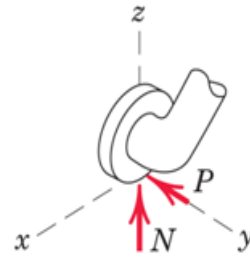
The procedure for solving three-dimensional rigid-body equilibrium problems is similar to that for two-dimensional problems. Whereas three governing equations were available for 2-D problems, we now have the following six governing equations for 3-D problems:

$$\Sigma F_x = 0 \quad (1) \qquad \Sigma M_x = 0 \quad (4)$$

$$\Sigma F_y = 0 \quad (2) \qquad \Sigma M_y = 0 \quad (5)$$

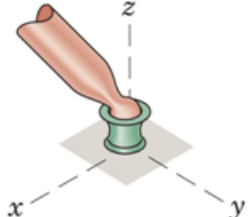
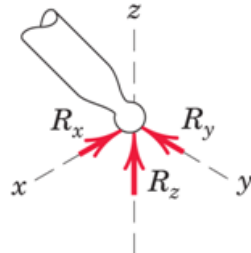
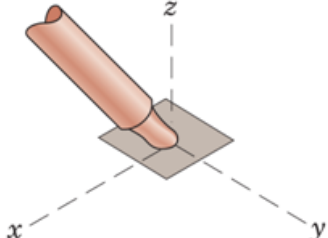
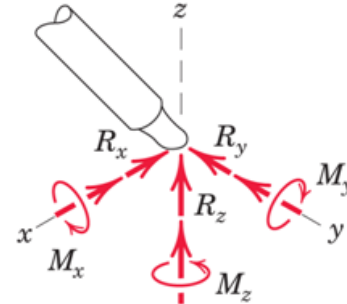
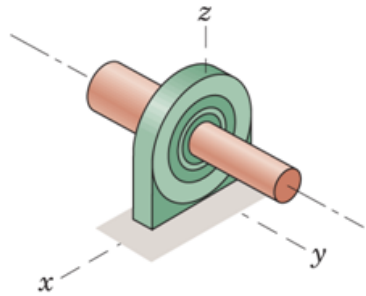
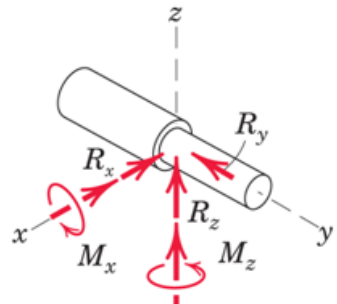
$$\Sigma F_z = 0 \quad (3) \qquad \Sigma M_z = 0 \quad (6)$$

MODELLING THE ACTION OF FORCES IN THREE DIMENSIONAL ANALYSIS

TYPE CONTACT AND FORCE ORIGIN	ACTION ON BODY TO BE ISOLATED
1. MEMBER IN CONTACT WITH SMOOTH SURFACE, OR BALL-SUPPORTED MEMBER 	 FORCE MUST BE NORMAL TO THE SURFACE AND DIRECTED TOWARD THE MEMBER
2. MEMBER IN CONTACT WITH ROUGH SURFACE 	 THE POSSIBILITY EXISTS FOR A FORCE F TANGENT TO THE SURFACE (FRICTION FORCE) TO ACT ON THE MEMBER, AS WELL AS A NORMAL FORCE N
3. ROLLER OR WHEEL SUPPORT WITH LATERAL CONSTRAINT 	 A LATERAL FORCE P EXERTED BY THE GUIDE ON THE WHEEL CAN EXIST IN ADDITION TO THE NORMAL FORCE N



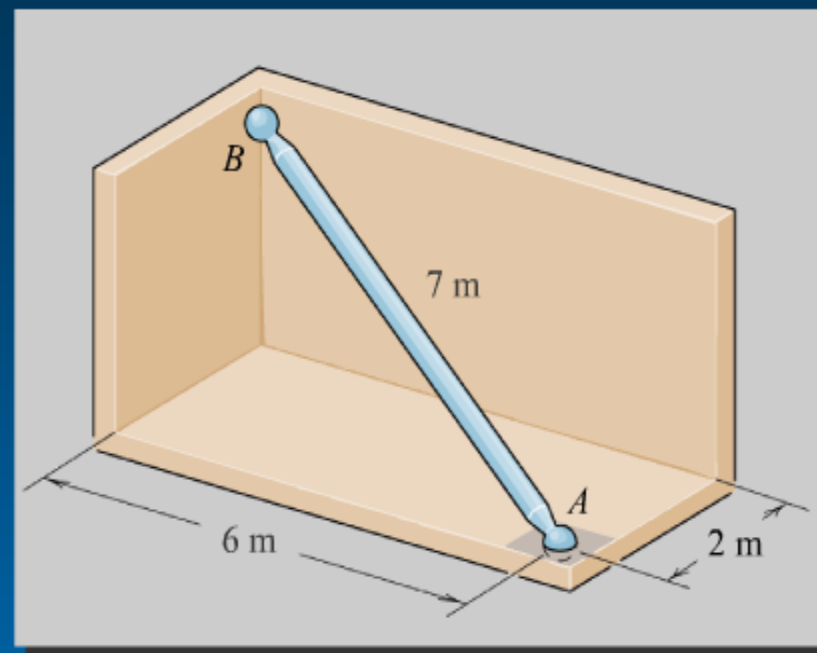
MODELLING THE ACTION OF FORCES IN THREE DIMENSIONAL ANALYSIS

TYPE CONTACT AND FORCE ORIGIN	ACTION ON BODY TO BE ISOLATED
4. BALL AND SOCKET JOINT 	 A BALL-AND-SOCKET JOINT FREE TO PIVOT ABOUT CENTER OF THE BALL CAN SUPPORT A FORCE R WITH ALL THREE COMPONENTS
5. FIXED CONNECTION (EMBEDDED OR WELDED) 	 IN ADDITION TO THREE COMPONENTS OF FORCE A FIXED CONNECTION CAN SUPPORT A COUPLE M REPRESENTED BY ITS THREE COMPONENTS
6. THRUST-BEARING SUPPORT 	 THRUST BEARING IS CAPABLE OF SUPPORTING AXIAL FORCE R_y, AS WELL AS RADIAL FORCES R_x AND R_z, COUPLES M_x AND M_z MUST IN SOME CASES BE ASSUMED ZERO IN ORDER TO PROVIDE STATICAL DETERMINACY.



Example

The uniform shaft with a mass of 200 kg is supported by a ball-and-socket joint at A . The ball end B rests against the smooth vertical walls. Determine the forces exerted by the walls and the floor on the ends of the shaft.

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Click "Next" to show the FBD



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Scalar Solution:

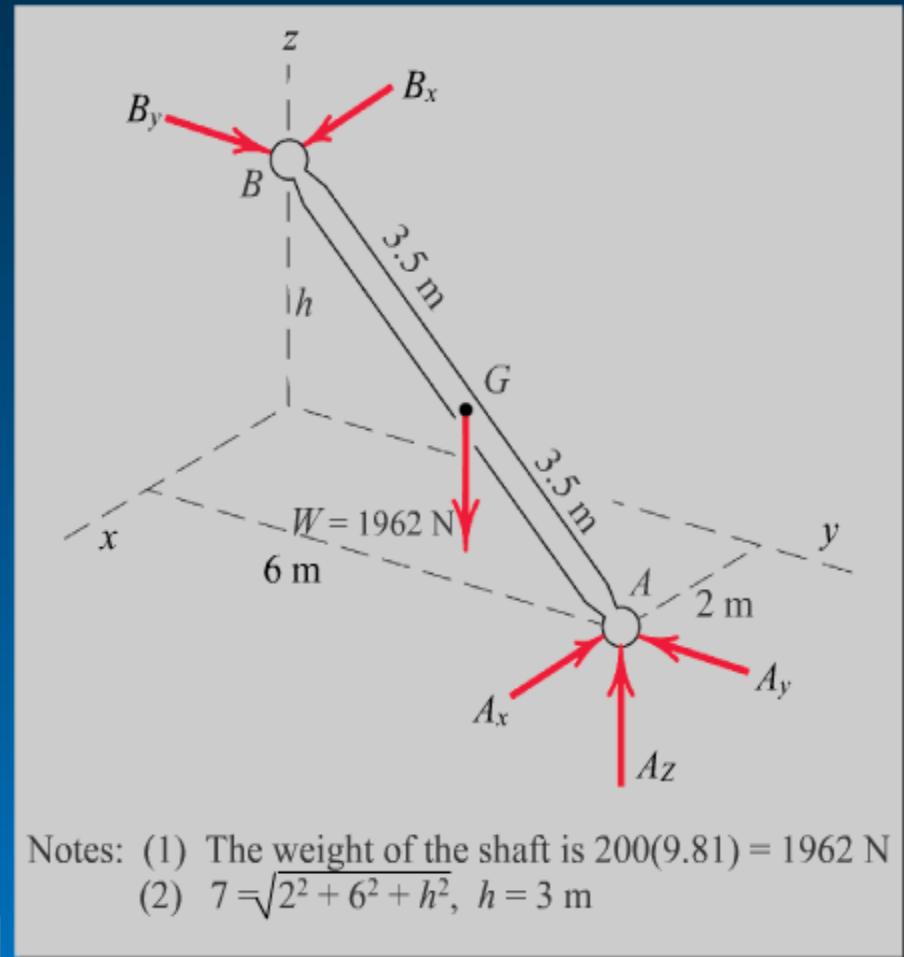
$$\Sigma F_x = 0 :$$

$$\Sigma F_y = 0 :$$

$$\Sigma F_z = 0 :$$

$$\Sigma M_{Ax} = 0 :$$

$$\Sigma M_{Ay} = 0 :$$



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Scalar Solution:

$$\Sigma F_x = 0 :$$

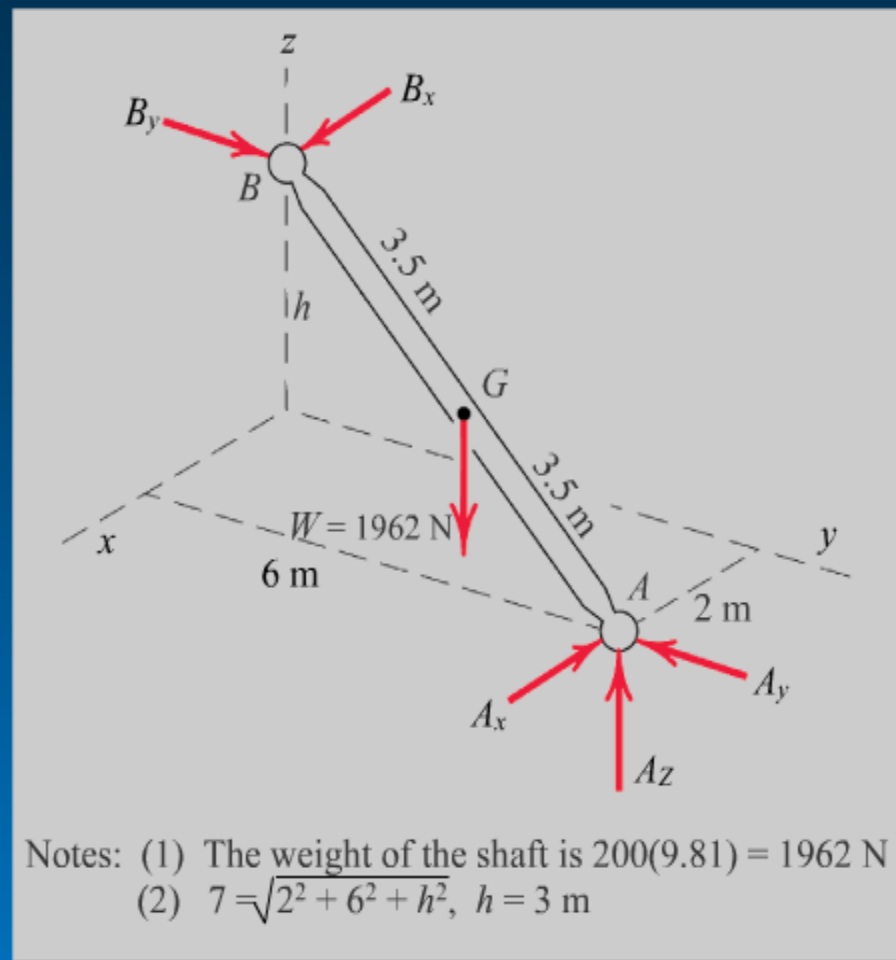
$$-A_x + B_x = 0$$

$$\Sigma F_y = 0 :$$

$$\Sigma F_z = 0 :$$

$$\Sigma M_{Ax} = 0 :$$

$$\Sigma M_{Ay} = 0 :$$



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Next



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Scalar Solution:

$$\Sigma F_x = 0 :$$

$$-A_x + B_x = 0$$

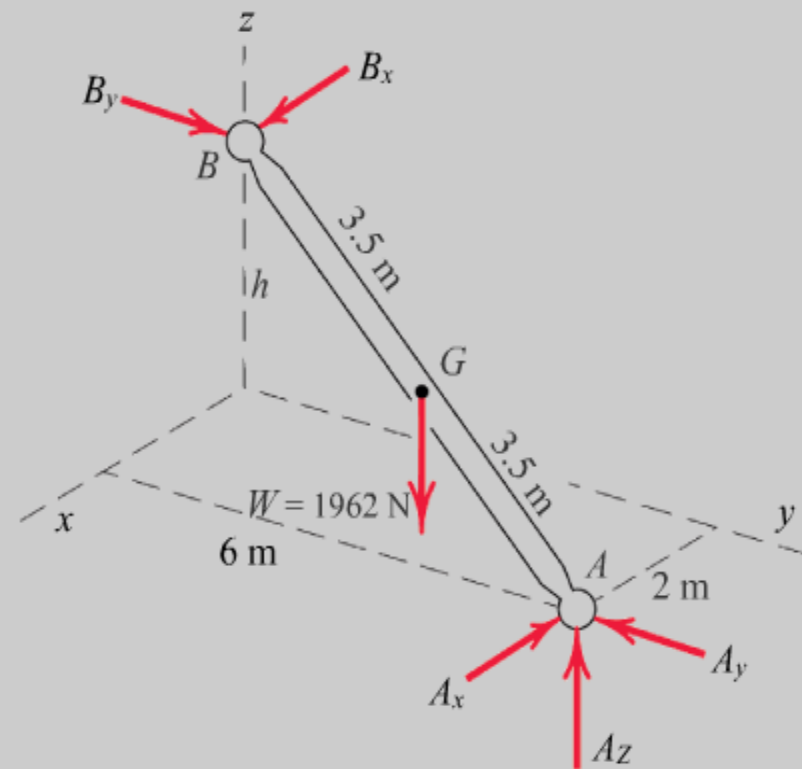
$$\Sigma F_y = 0 :$$

$$-A_y + B_y = 0$$

$$\Sigma F_z = 0 :$$

$$\Sigma M_{Ax} = 0 :$$

$$\Sigma M_{Ay} = 0 :$$



Notes: (1) The weight of the shaft is $200(9.81) = 1962 \text{ N}$

(2) $7 = \sqrt{2^2 + 6^2 + h^2}$, $h = 3 \text{ m}$

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Scalar Solution:

$$\Sigma F_x = 0 :$$

$$-A_x + B_x = 0$$

$$\Sigma F_y = 0 :$$

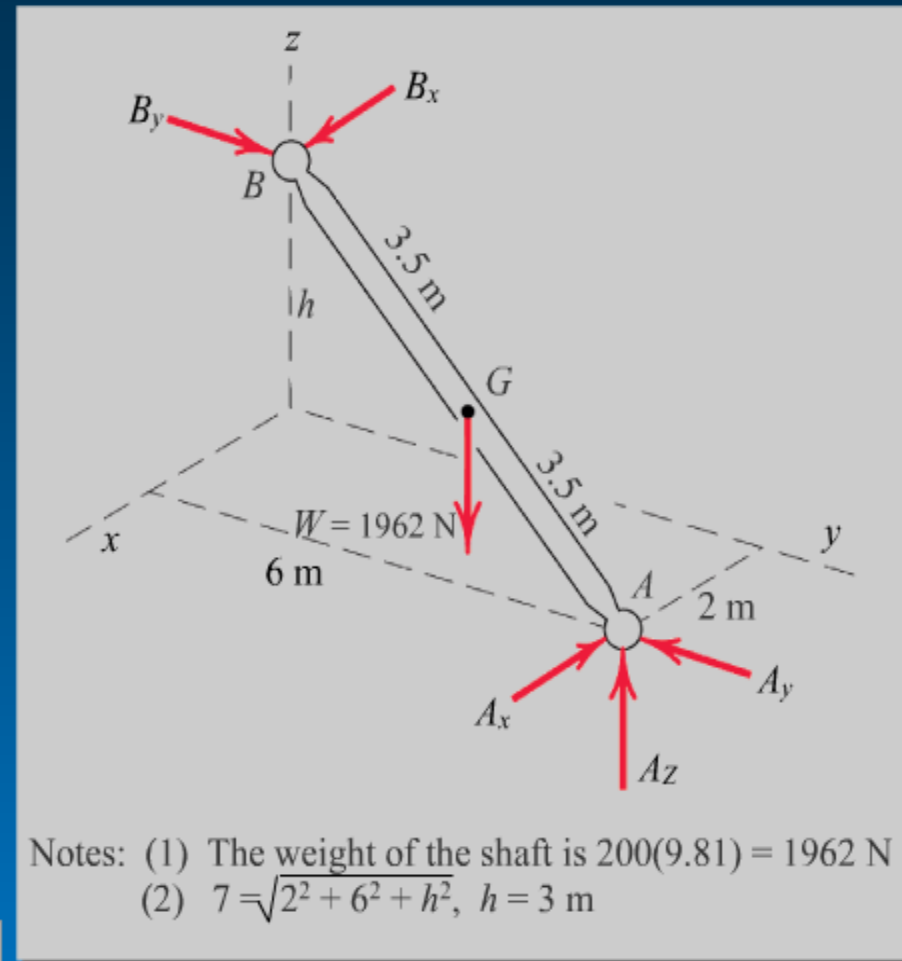
$$-A_y + B_y = 0$$

$$\Sigma F_z = 0 :$$

$$A_z - 1962 = 0$$

$$\Sigma M_{Ax} = 0 :$$

$$\Sigma M_{Ay} = 0 :$$



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Scalar Solution:

$$\Sigma F_x = 0 :$$

$$-A_x + B_x = 0$$

$$\Sigma F_y = 0 :$$

$$-A_y + B_y = 0$$

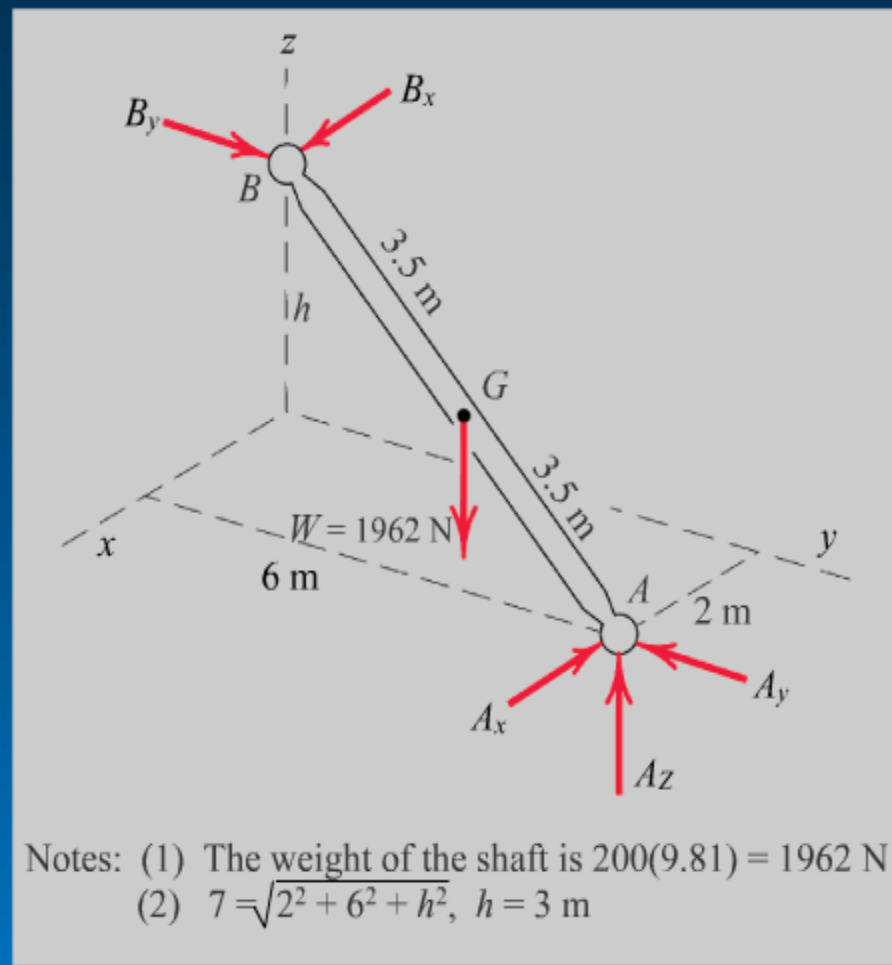
$$\Sigma F_z = 0 :$$

$$A_z - 1962 = 0$$

$$\Sigma M_{Ax} = 0 :$$

$$1962(3) - 3B_y = 0$$

$$\Sigma M_{Ay} = 0 :$$



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Scalar Solution:

$$\Sigma F_x = 0 :$$

$$-A_x + B_x = 0$$

$$\Sigma F_y = 0 :$$

$$-A_y + B_y = 0$$

$$\Sigma F_z = 0 :$$

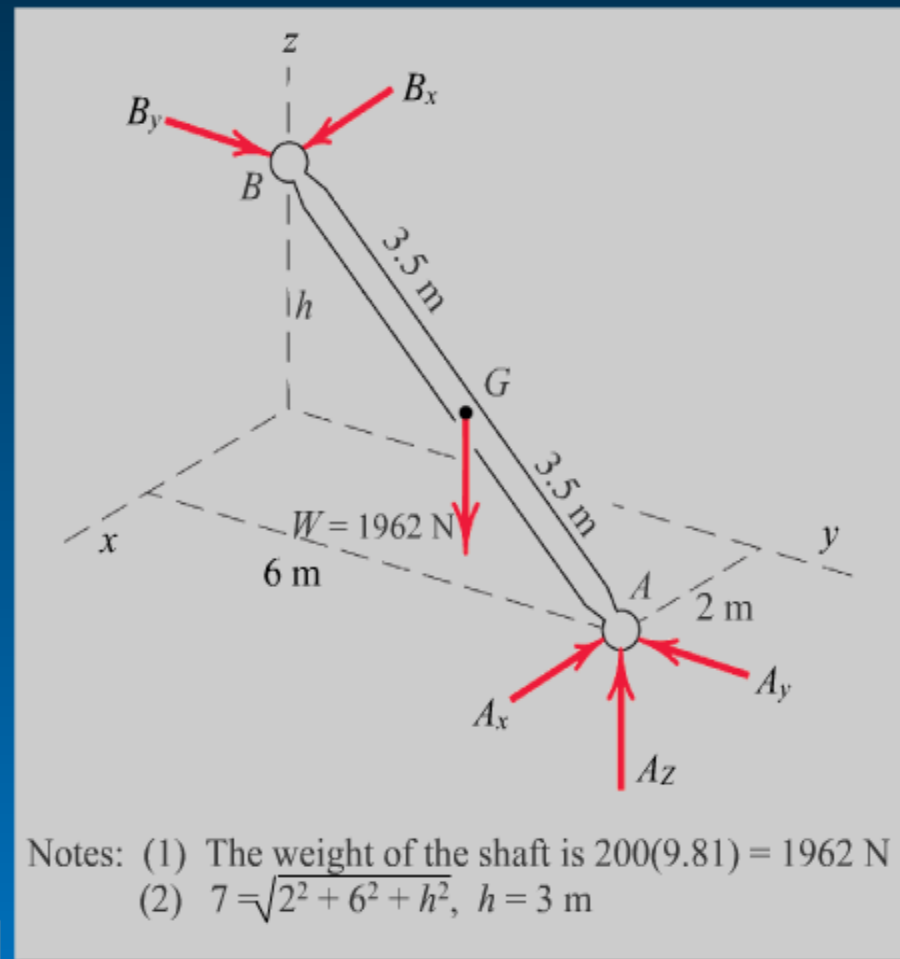
$$A_z - 1962 = 0$$

$$\Sigma M_{Ax} = 0 :$$

$$1962(3) - 3B_y = 0$$

$$\Sigma M_{Ay} = 0 :$$

$$-1962(1) + 3B_x = 0$$



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Scalar Solution:

$$\Sigma F_x = 0 :$$

$$-A_x + B_x = 0$$

$$\Sigma F_y = 0 :$$

$$-A_y + B_y = 0$$

$$\Sigma F_z = 0 :$$

$$A_z - 1962 = 0$$

$$\Sigma M_{Ax} = 0 :$$

$$1962(3) - 3B_y = 0$$

$$\Sigma M_{Ay} = 0 :$$

$$-1962(1) + 3B_x = 0$$

Solution

$$A_x = 654 \text{ N} \quad B_x = 654 \text{ N}$$

$$A_y = 1962 \text{ N} \quad B_y = 1962 \text{ N}$$

$$A_z = 1962 \text{ N}$$

The magnitude of the force at A is:

$$A = \sqrt{A_x^2 + A_y^2 + A_z^2} = 2850 \text{ N}$$

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Comments:

Since we have only 5 unknowns, we need only 5 equations in total to solve for these unknowns.

The equation $\Sigma M_{Az} = 0$ can be used as a check.

For this example problem, we have

$$\Sigma M_{Az} = B_x(6) - B_y(2) = 654(6) - 1962(2) = 0$$

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Click "Back" to go back to solution



4/6



Sample Problem 3/2

Calculate the tension T in the cable which supports the 500-kg mass with the pulley arrangement shown. Each pulley is free to rotate about its bearing, and the weights of all parts are small compared with the load. Find the magnitude of the total force on the bearing of pulley C .

Solution. The free-body diagram of each pulley is drawn in its relative position to the others. We begin with pulley A , which includes the only known force. With the unspecified pulley radius designated by r , the equilibrium of moments about its center O and the equilibrium of forces in the vertical direction require

$$\begin{aligned} \textcircled{1} \quad [\Sigma M_O = 0] \quad T_1 r - T_2 r = 0 \quad T_1 = T_2 \\ [\Sigma F_y = 0] \quad T_1 + T_2 - 500(9.81) = 0 \quad 2T_1 = 500(9.81) \quad T_1 = T_2 = 2450 \text{ N} \end{aligned}$$

From the example of pulley A we may write the equilibrium of forces on pulley B by inspection as

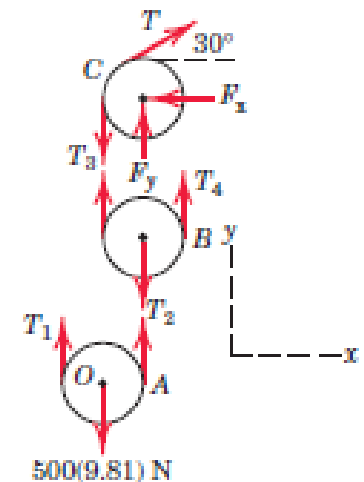
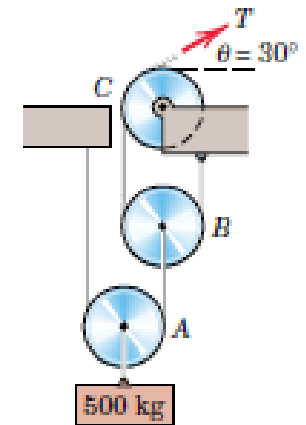
$$T_3 = T_4 = T_2/2 = 1226 \text{ N}$$

For pulley C the angle $\theta = 30^\circ$ in no way affects the moment of T about the center of the pulley, so that moment equilibrium requires

$$T = T_3 \quad \text{or} \quad T = 1226 \text{ N} \quad \text{Ans.}$$

Equilibrium of the pulley in the x - and y -directions requires

$$\begin{aligned} [\Sigma F_x = 0] \quad 1226 \cos 30^\circ - F_x = 0 \quad F_x = 1062 \text{ N} \\ [\Sigma F_y = 0] \quad F_y + 1226 \sin 30^\circ - 1226 = 0 \quad F_y = 613 \text{ N} \\ [F = \sqrt{F_x^2 + F_y^2}] \quad F = \sqrt{(1062)^2 + (613)^2} = 1226 \text{ N} \quad \text{Ans.} \end{aligned}$$



Helpful Hint

- ① Clearly the radius r does not influence the results. Once we have analyzed a simple pulley, the results should be perfectly clear by inspection.

Sample Problem 3/3

The uniform 100-kg I-beam is supported initially by its end rollers on the horizontal surface at A and B . By means of the cable at C it is desired to elevate end B to a position 3 m above end A . Determine the required tension P , the reaction at A , and the angle θ made by the beam with the horizontal in the elevated position.

Solution. In constructing the free-body diagram, we note that the reaction on the roller at A and the weight are vertical forces. Consequently, in the absence of other horizontal forces, P must also be vertical. From Sample Problem 3/2 we see immediately that the tension P in the cable equals the tension P applied to the beam at C .

Moment equilibrium about A eliminates force R and gives

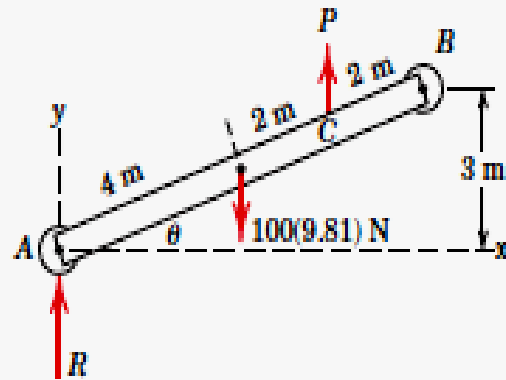
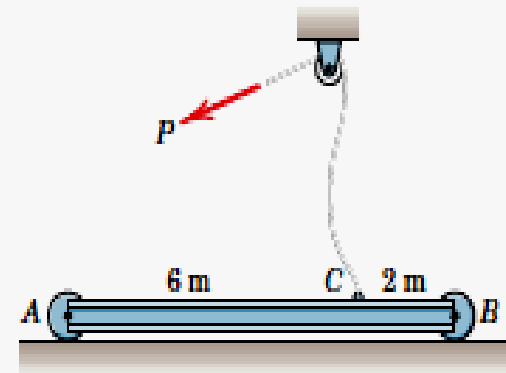
$$\textcircled{1} [\Sigma M_A = 0] \quad P(6 \cos \theta) - 981(4 \cos \theta) = 0 \quad P = 654 \text{ N} \quad \text{Ans.}$$

Equilibrium of vertical forces requires

$$[\Sigma F_y = 0] \quad 654 + R - 981 = 0 \quad R = 327 \text{ N} \quad \text{Ans.}$$

The angle θ depends only on the specified geometry and is

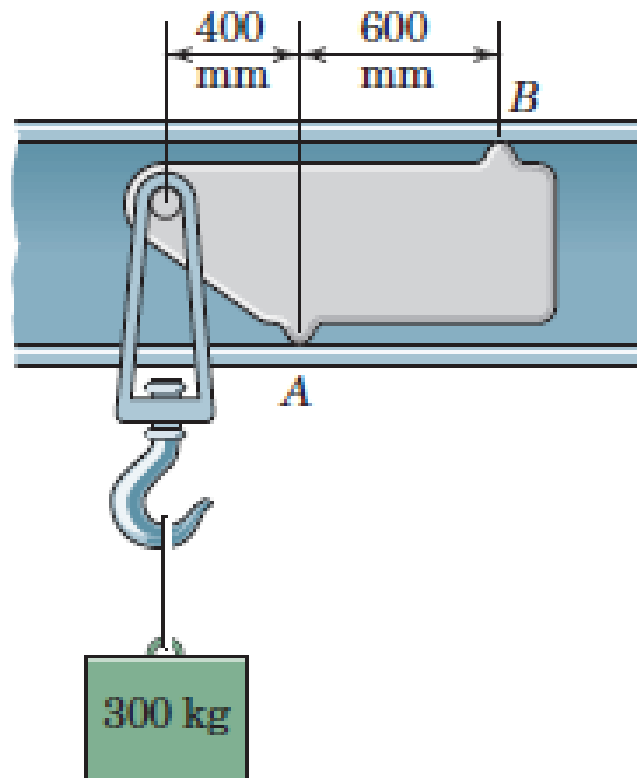
$$\sin \theta = 3/8 \quad \theta = 22.0^\circ \quad \text{Ans.}$$



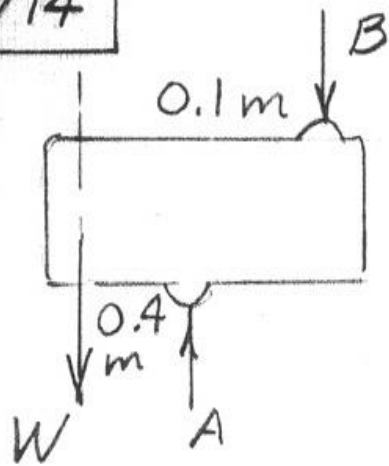
Helpful Hint

- ① Clearly the equilibrium of this parallel force system is independent of θ .

3/14 To facilitate shifting the position of a lifting hook when it is not under load, the sliding hanger shown is used. The projections at A and B engage the flanges of a box beam when a load is supported, and the hook projects through a horizontal slot in the beam. Compute the forces at A and B when the hook supports a 300-kg mass.



3/14



$$W = 300(9.81) = 2943 \text{ N}$$

$$\sum M_A = 0; 2943(0.4) - B(0.6) = 0$$

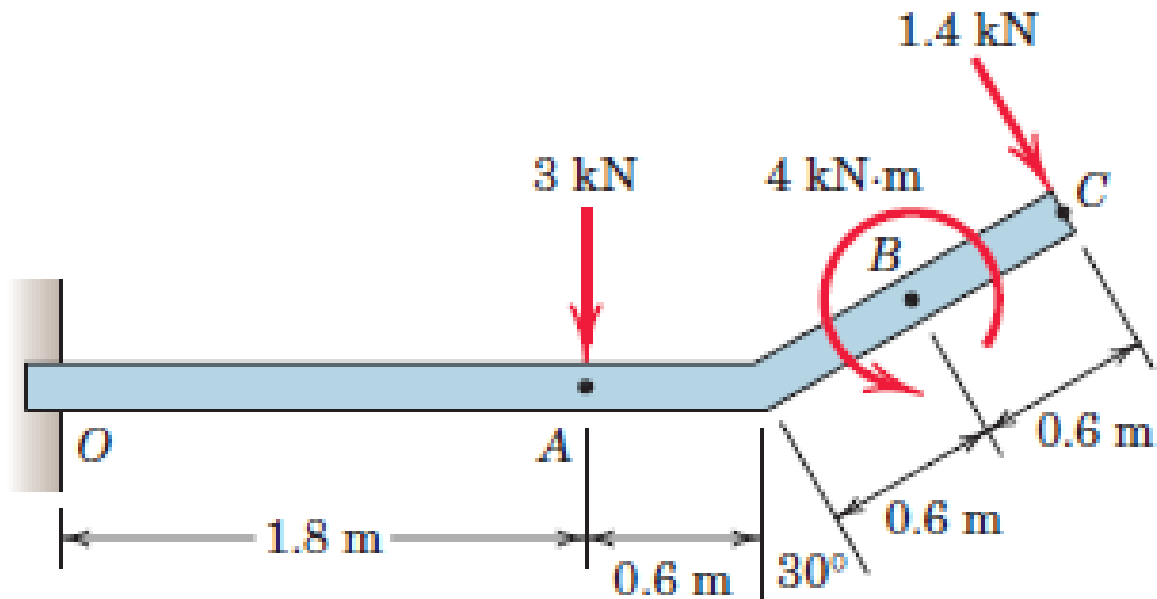
$$B = 1962 \text{ N or } \underline{1.962 \text{ kN}}$$

$$\sum F = 0; A = 2943 + 1962$$

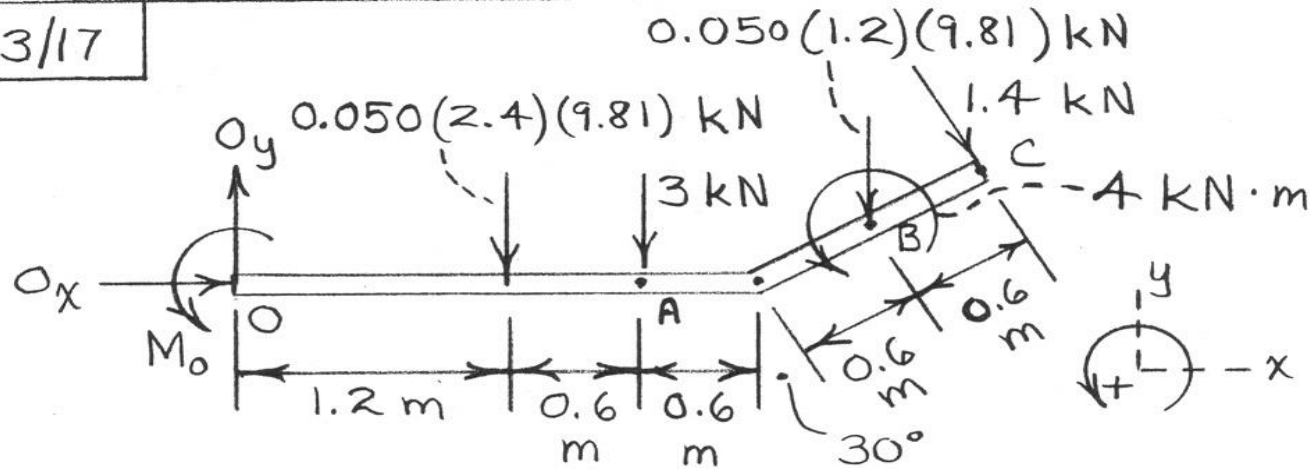
$$= \underline{4910 \text{ N or } 4.91 \text{ kN}}$$

3/17 The uniform beam has a mass of 50 kg per meter of length. Compute the reactions at the support O . The force loads shown lie in a vertical plane.

Ans. $O_x = -0.7 \text{ kN}$, $O_y = 5.98 \text{ kN}$, $M_O = 9.12 \text{ kN} \cdot \text{m}$



3/17



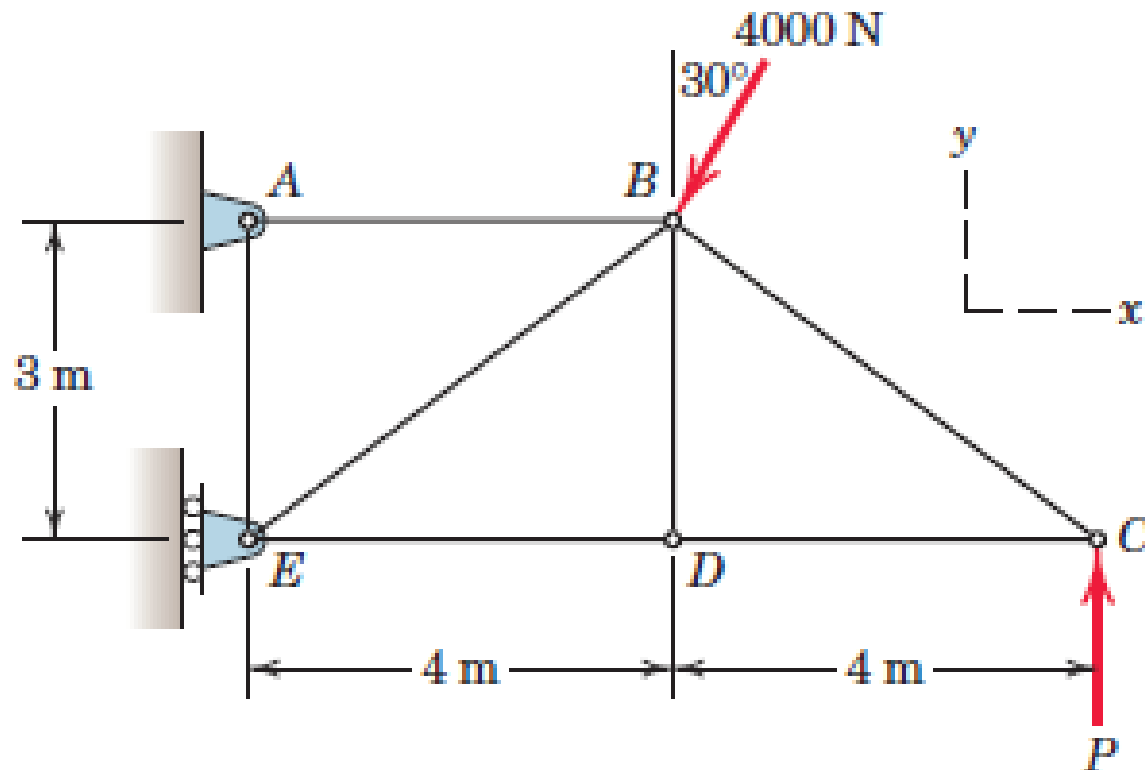
$$\sum F_x = 0 : O_x + 1.4 \sin 30^\circ = 0$$

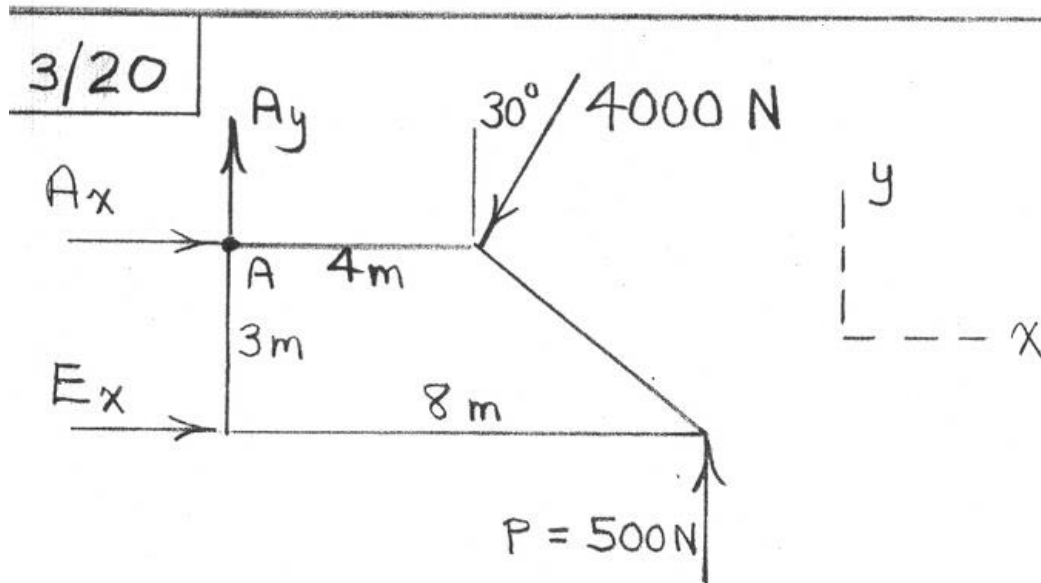
$$O_x = -0.7 \text{ kN}$$

$$\sum F_y = 0 : O_y - 0.050(2.4)(9.81) - 3 - 1.4 \cos 30^\circ - 0.050(1.2)(9.81) = 0, \quad O_y = 5.98 \text{ kN}$$

$$\sum M_o = 0 : M_o - 0.050(2.4)(9.81)(1.2) - 3(1.8) - 0.050(1.2)(9.81)(2.4 + 0.6 \cos 30^\circ) + 4 - 1.4(2.4 \cos 30^\circ + 1.2) = 0, \quad M_o = 9.12 \text{ kN}\cdot\text{m}$$

- 3/20** Determine the reactions at A and E if $P = 500$ N. What is the maximum value which P may have for static equilibrium? Neglect the weight of the structure compared with the applied loads.





$$\sum F_x = 0: A_x + E_x - 4000 \sin 30^\circ = 0$$

$$\sum F_y = 0: A_y - 4000 \cos 30^\circ + 500 = 0$$

$$\sum M_A = 0: E_x(3) + 500(8) - 4000 \cos 30^\circ(4) = 0$$

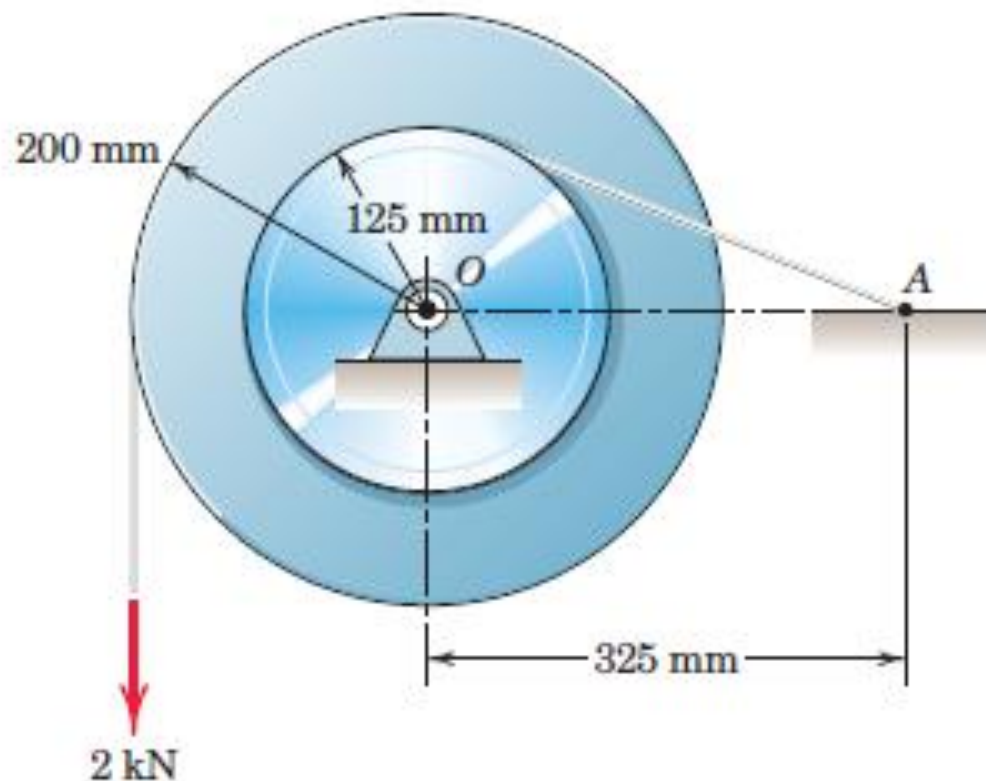
$$\Rightarrow \underline{A_x = -1290 \text{ N}}, \quad \underline{A_y = 2960 \text{ N}}, \quad \underline{E_x = 3290 \text{ N}}$$

For maximum P: $E_x = 0$ and $\sum M_A = 0:$

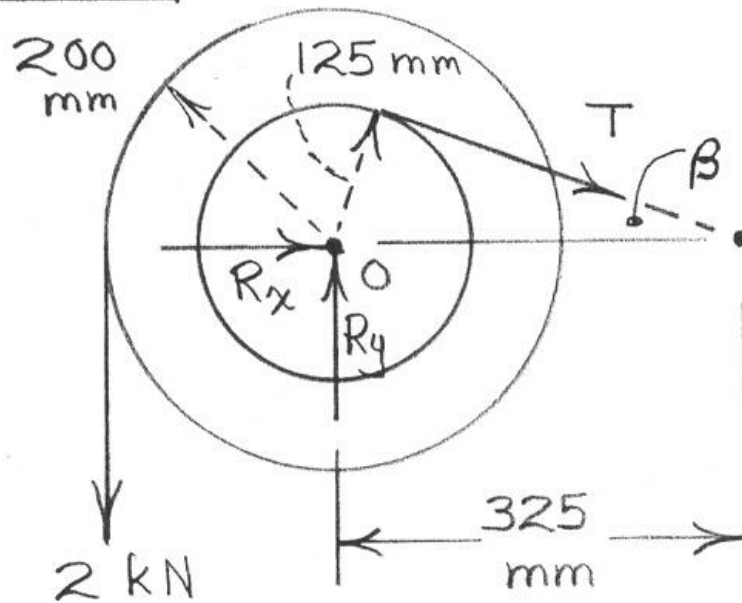
$$P(8) - 4000 \cos 30^\circ(4) = 0, \quad \underline{P = 1732 \text{ N}}$$

- 3/31** The two light pulleys are fastened together and form an integral unit. They are prevented from turning about their bearing at O by a cable wound securely around the smaller pulley and fastened to point A . Calculate the magnitude R of the force supported by the bearing O for the applied 2-kN load.

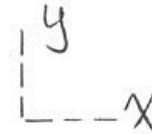
Ans. $R = 4.38$ kN



3/31



$$\begin{cases} \sin \beta = \frac{125}{325} = \frac{5}{13} \\ \cos \beta = \frac{12}{13} \end{cases}$$



$$\sum M_O = 0: 2(200) - T(125) = 0, T = 3.2 \text{ kN}$$

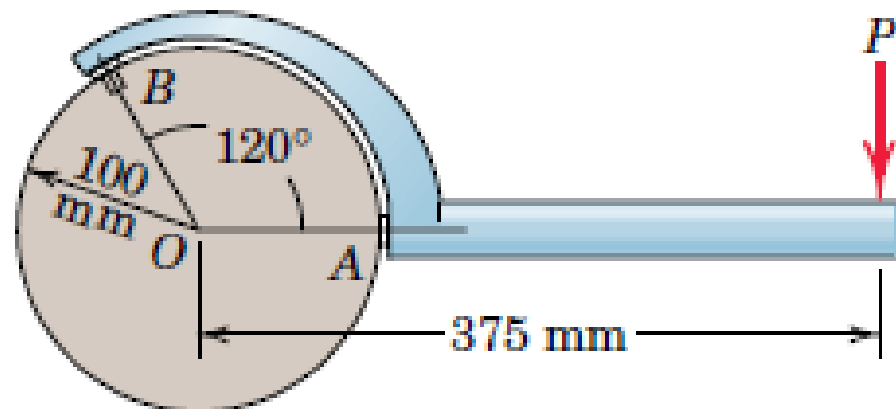
$$\sum F_x = 0: 3.2\left(\frac{12}{13}\right) + R_x = 0, R_x = -2.95 \text{ kN}$$

$$\sum F_y = 0: R_y - 2 - 3.2\left(\frac{5}{13}\right) = 0, R_y = 3.23 \text{ kN}$$

$$R = \sqrt{R_x^2 + R_y^2} = \sqrt{2.95^2 + 3.23^2} = \underline{4.38 \text{ kN}}$$

- 3/43** The hook wrench or pin spanner is used to turn shafts and collars. If a moment of $80 \text{ N}\cdot\text{m}$ is required to turn the 200-mm-diameter collar about its center O under the action of the applied force P , determine the contact force R on the smooth surface at A . Engagement of the pin at B may be considered to occur at the periphery of the collar.

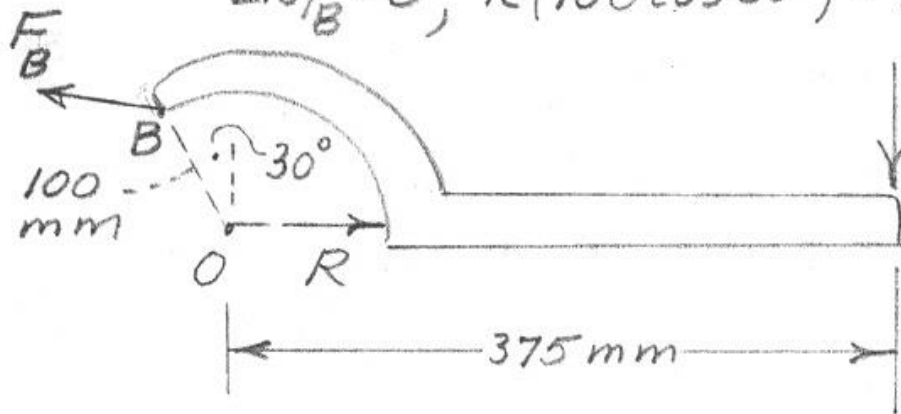
Ans. $R = 1047 \text{ N}$



3/43

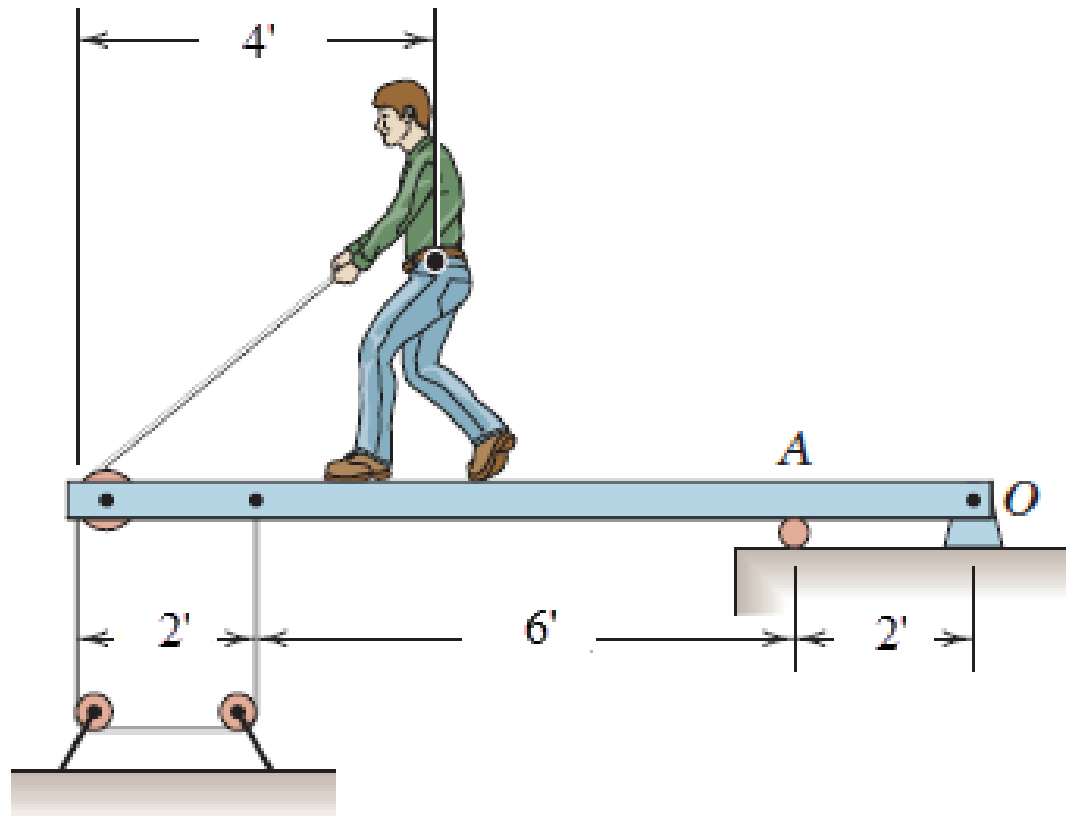
$$M = Fd; 80 = P(0.375), P = 213 \text{ N}$$

$$\sum M_B = 0; R(100 \cos 30^\circ) - (375 + 100 \sin 30^\circ)(213) = 0$$

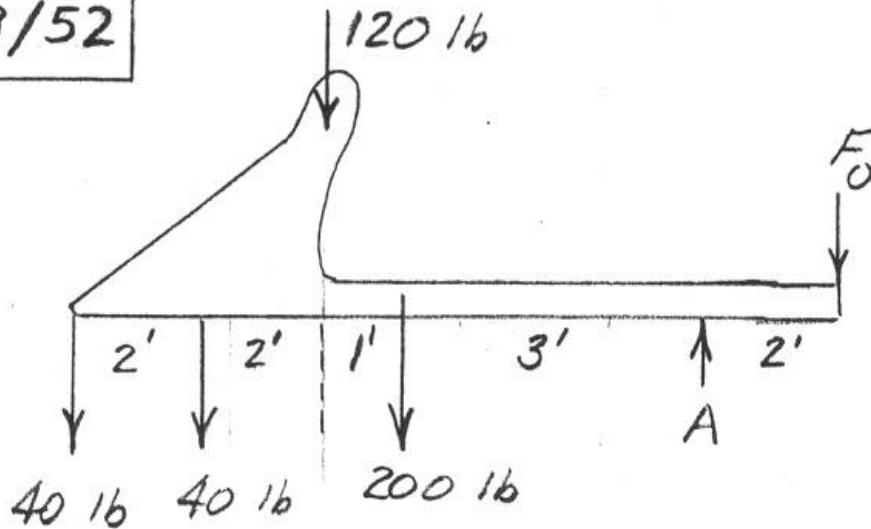


$$\underline{R = 1047 \text{ N}}$$

- 3/52** To test the deflection of the uniform 200 lb beam the 120 lb boy exerts a pull of 40 lb on the rope rigged as shown. Compute the force supported by the pin at the hinge O.



3/52



$$\sum M_A = 0;$$

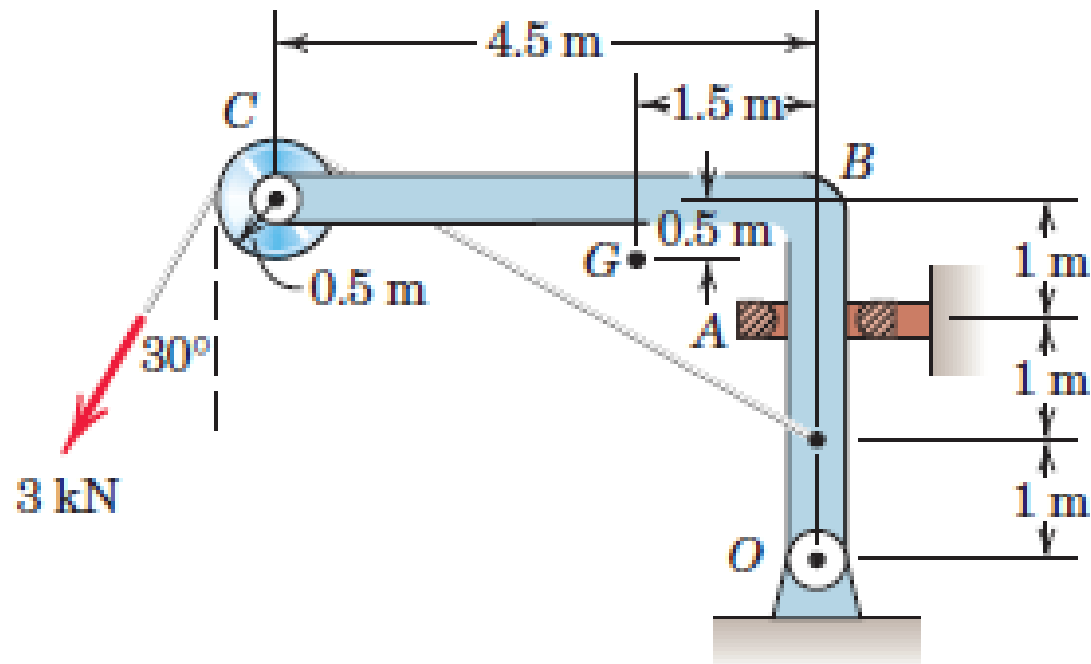
$$40(8) + 40(6)$$

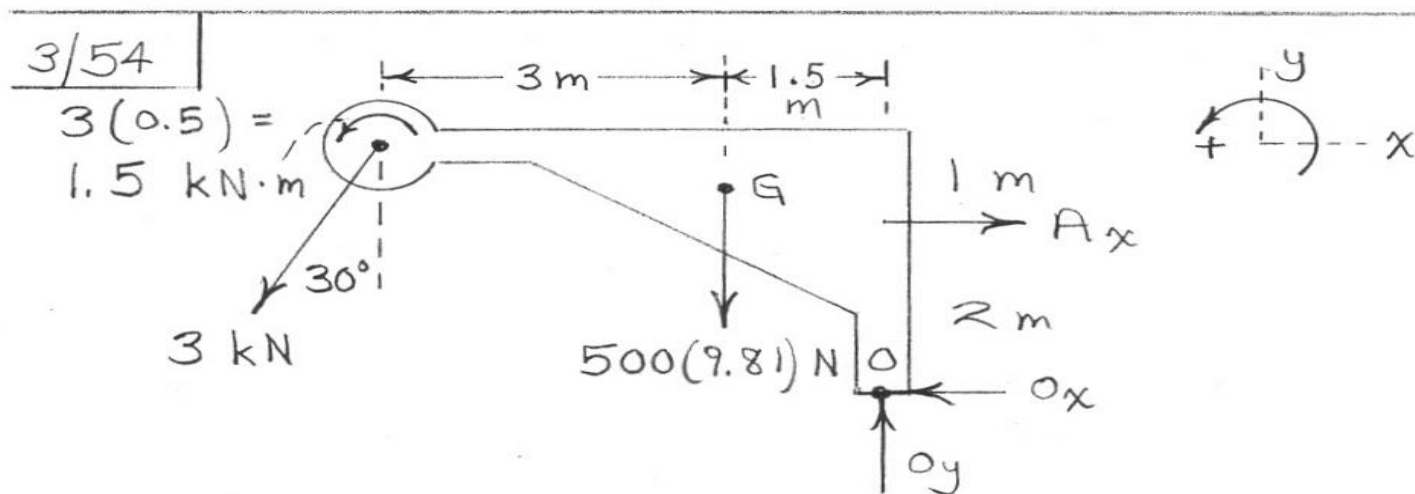
$$+ 120(4) + 200(3)$$

$$- 2F_0 = 0$$

$$F_0 = 820 \text{ lb}$$

- 3/54** The member OBC and sheave at C together have a mass of 500 kg, with a combined center of mass at G . Calculate the magnitude of the force supported by the pin connection at O when the 3-kN load is applied. The collar at A can provide support in the horizontal direction only.





$$\sum M_O = 0 : -2A + 500(9.81)(1.5) + 1500 + 3000 \cos 30^\circ (4.5) + 3000 \sin 30^\circ (3) = 0$$

$$A = 12520 \text{ N}$$

$$\sum F_x = 0 : -3000 \sin 30^\circ + 12520 - O_x = 0$$

$$O_x = 11020 \text{ N}$$

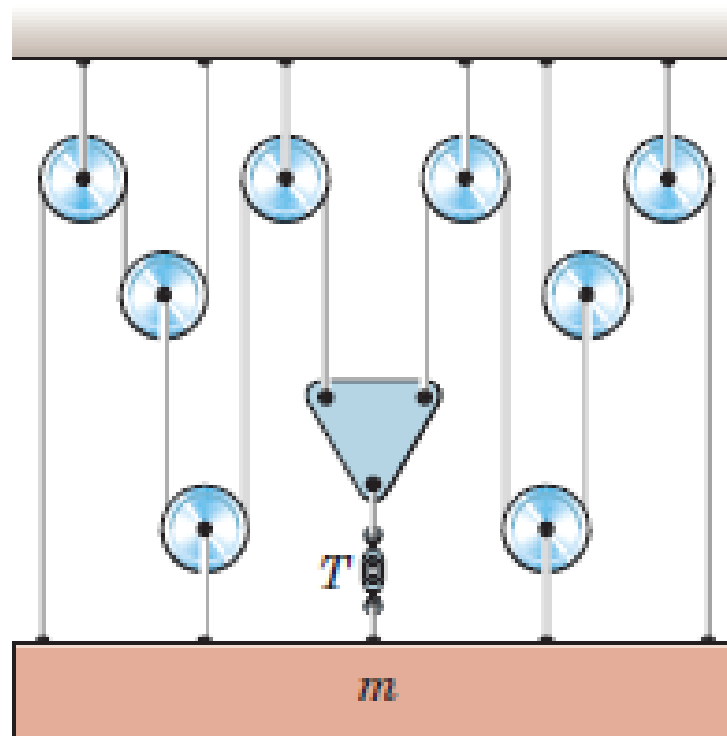
$$\sum F_y = 0 : -3000 \cos 30^\circ - 500(9.81) + O_y = 0$$

$$O_y = 7500 \text{ N}$$

$$O = \sqrt{11020^2 + 7500^2} = 13340 \text{ N or } \underline{13.34 \text{ kN}}$$

3/57 Determine the tension T in the turnbuckle for the pulley-cable system in terms of the mass m of the body which it supports. Neglect the mass of the pulleys and cable.

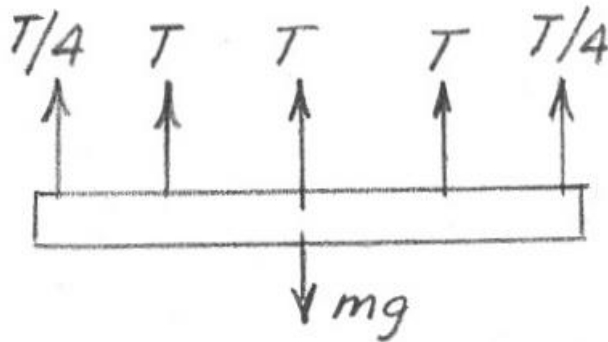
Ans. $T = \frac{2}{7}mg$



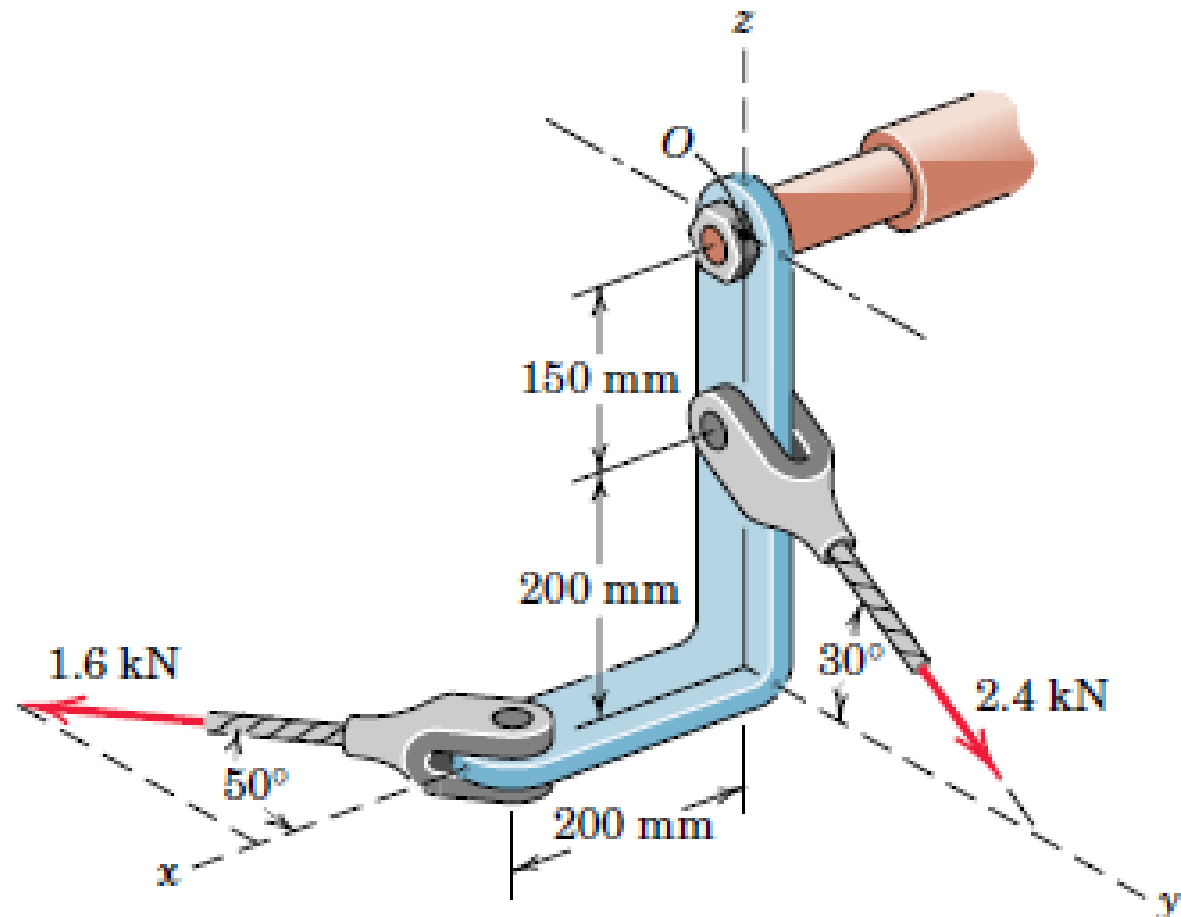
3/57

$$\Sigma F = 0; \left(3 + \frac{1}{4} + \frac{1}{4}\right)T - mg = 0$$

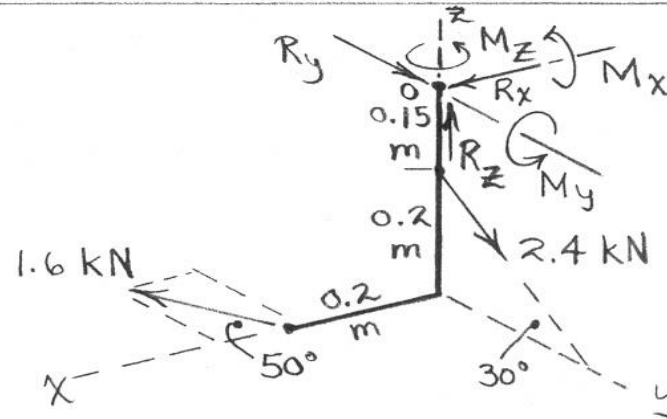
$$T = \frac{2}{7}mg$$



3/78 Determine the magnitudes of the force **R** and couple **M** exerted by the nut and bolt on the loaded bracket at **O** to maintain equilibrium.



3/78



$$\sum F_x = 0 : R_x + 1.6 \cos 50^\circ, \quad R_x = -1.028 \text{ kN}$$

$$\sum F_y = 0 : R_y + 2.4 \cos 30^\circ - 1.6 \sin 50^\circ = 0, \quad R_y = -0.853 \text{ kN}$$

$$\sum F_z = 0 : R_z - 2.4 \sin 30^\circ = 0, \quad R_z = 1.2 \text{ kN}$$

$$R = \sqrt{R_x^2 + R_y^2 + R_z^2} = \underline{1.796 \text{ kN}}$$

$$\sum M_{O_x} = 0 : M_x + 2.4 \cos 30^\circ (0.15) - 1.6 \sin 50^\circ (0.35) = 0$$

$$M_x = 0.1172 \text{ kN}\cdot\text{m}$$

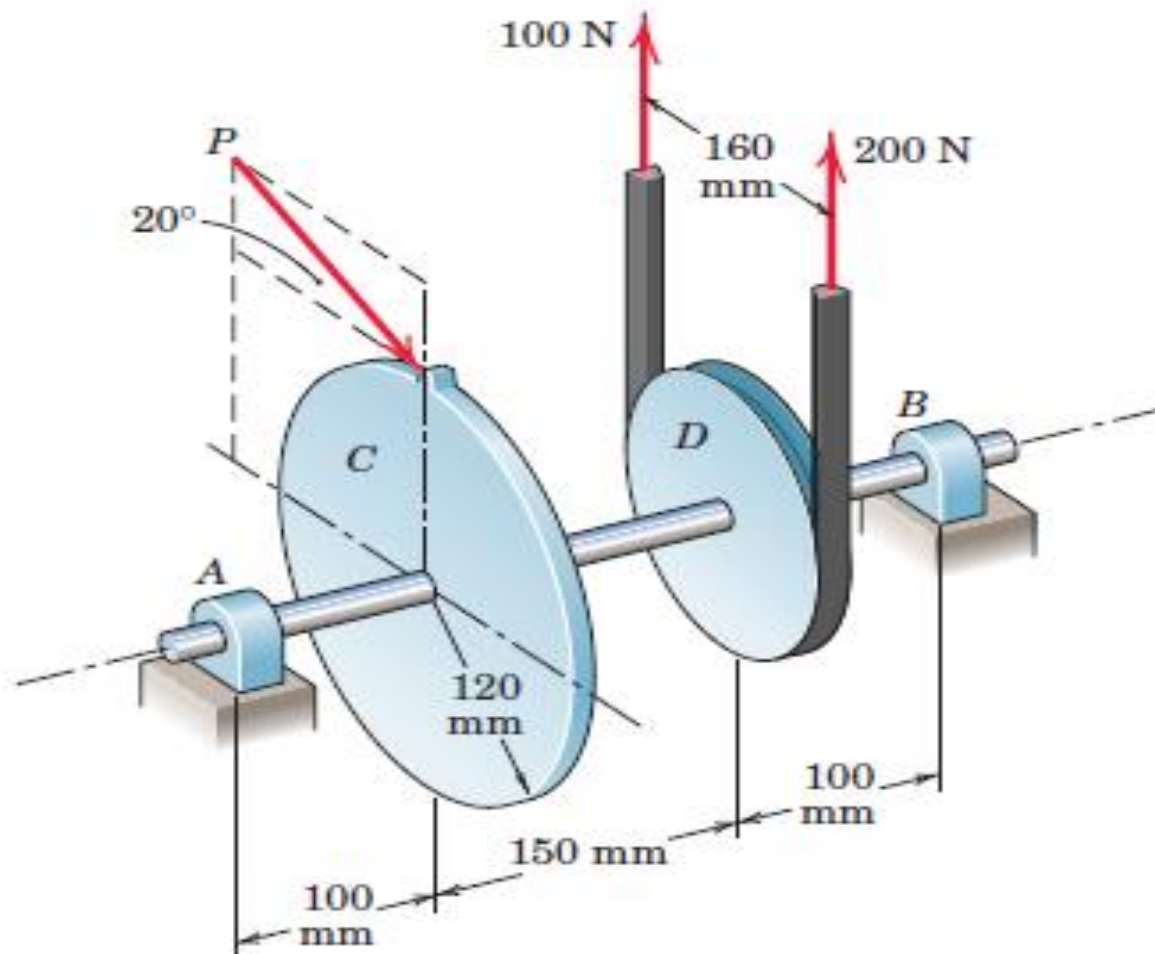
$$\sum M_{O_y} = 0 : M_y - 1.6 \cos 50^\circ (0.35) = 0, \quad M_y = 0.360 \text{ kN}\cdot\text{m}$$

$$\sum M_{O_z} = 0 : M_z - 1.6 \sin 50^\circ (0.2) = 0, \quad M_z = 0.245 \text{ kN}\cdot\text{m}$$

$$M = \sqrt{M_x^2 + M_y^2 + M_z^2} = \underline{0.451 \text{ kN}\cdot\text{m}}$$

3/83 Gear *C* drives the V-belt pulley *D* at a constant speed. For the belt tensions shown calculate the gear-tooth force *P* and the magnitudes of the total forces supported by the bearings at *A* and *B*.

Ans. $P = 70.9 \text{ N}$, $A = 83.3 \text{ N}$, $B = 208 \text{ N}$



3/83

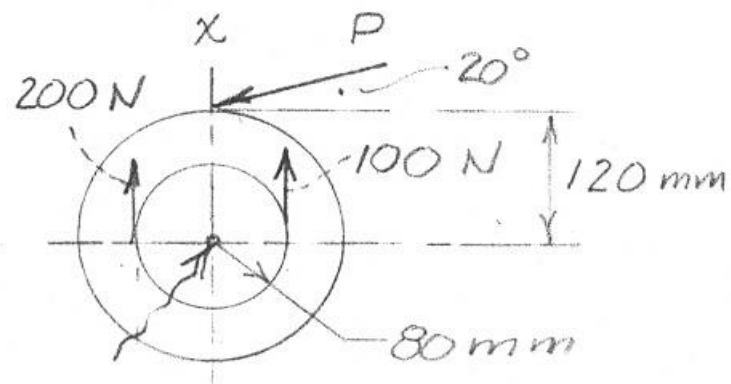
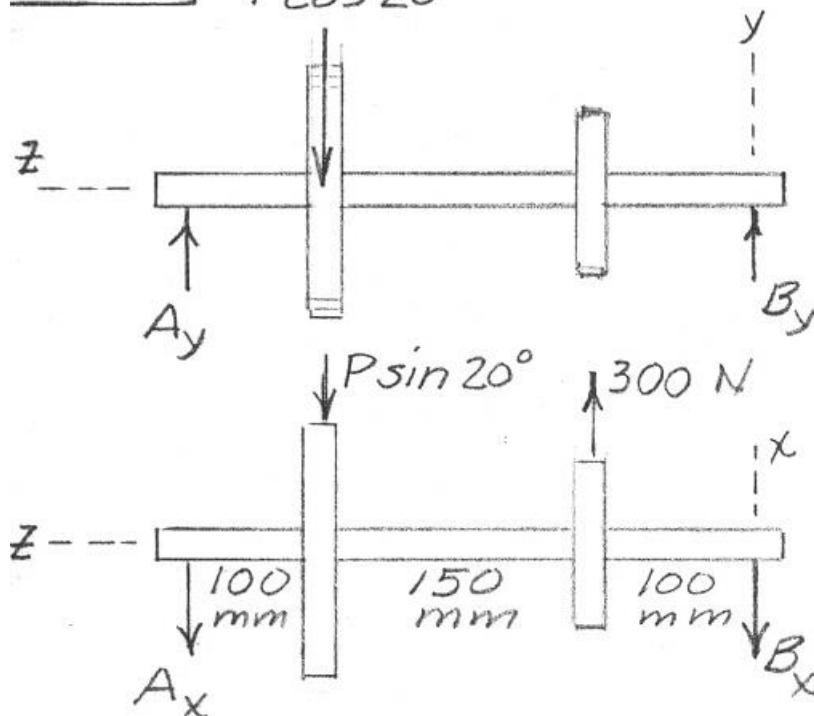
 $P \cos 20^\circ$

$$\Sigma M_z = 0;$$

$$(200 - 100) 0.080$$

$$- 0.120 P \cos 20^\circ = 0$$

$$P = 70.9 \text{ N}$$



$$y-z; \Sigma M_A = 0; 0.35 B_y - 70.9 \cos 20^\circ (0.1) = 0, B_y = 19.05 \text{ N}$$

$$\Sigma F_y = 0; A_y + 19.05 - 70.9 \cos 20^\circ = 0, A_y = 47.6 \text{ N}$$

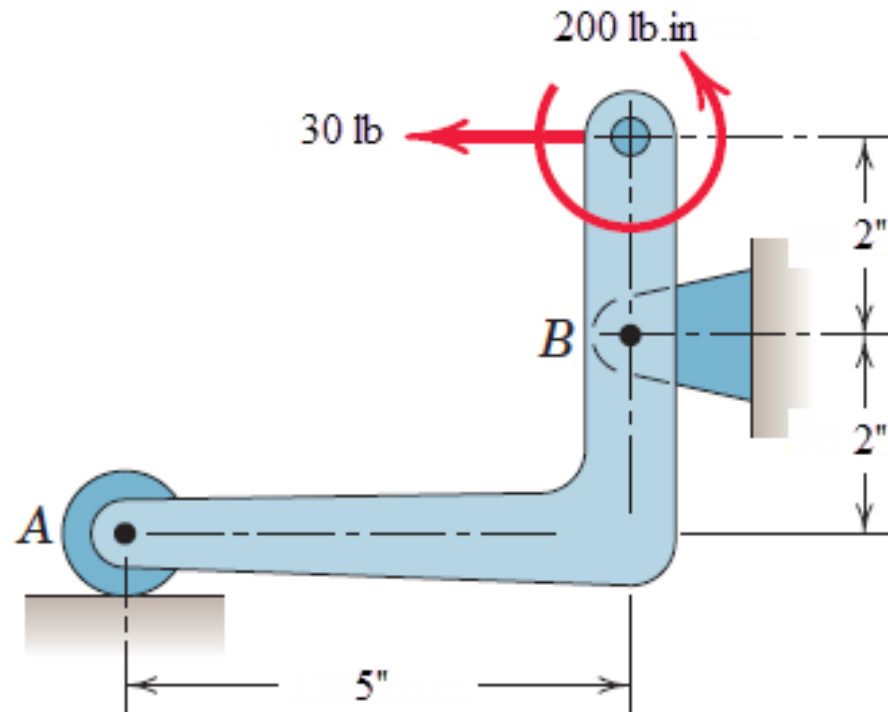
$$x-z; \Sigma M_A = 0; 300(0.25) - 70.9 \sin 20^\circ (0.1) - 0.35 B_x = 0$$

$$B_x = 207 \text{ N}$$

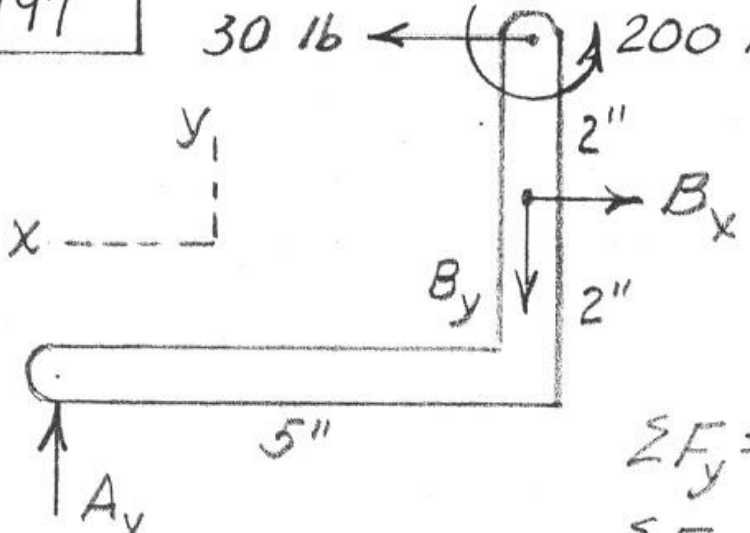
$$\Sigma F_x = 0; A_x + 70.9 \sin 20^\circ + 207 - 300 = 0; A_x = 68.4 \text{ N}$$

$$A = \sqrt{68.4^2 + 47.6^2} = 83.3 \text{ N}, B = \sqrt{207^2 + 19.05^2} = 208 \text{ N}$$

3/97 Calculate the magnitude of the force supported by the pin at B for the bell crank loaded and supported as shown.



3/97

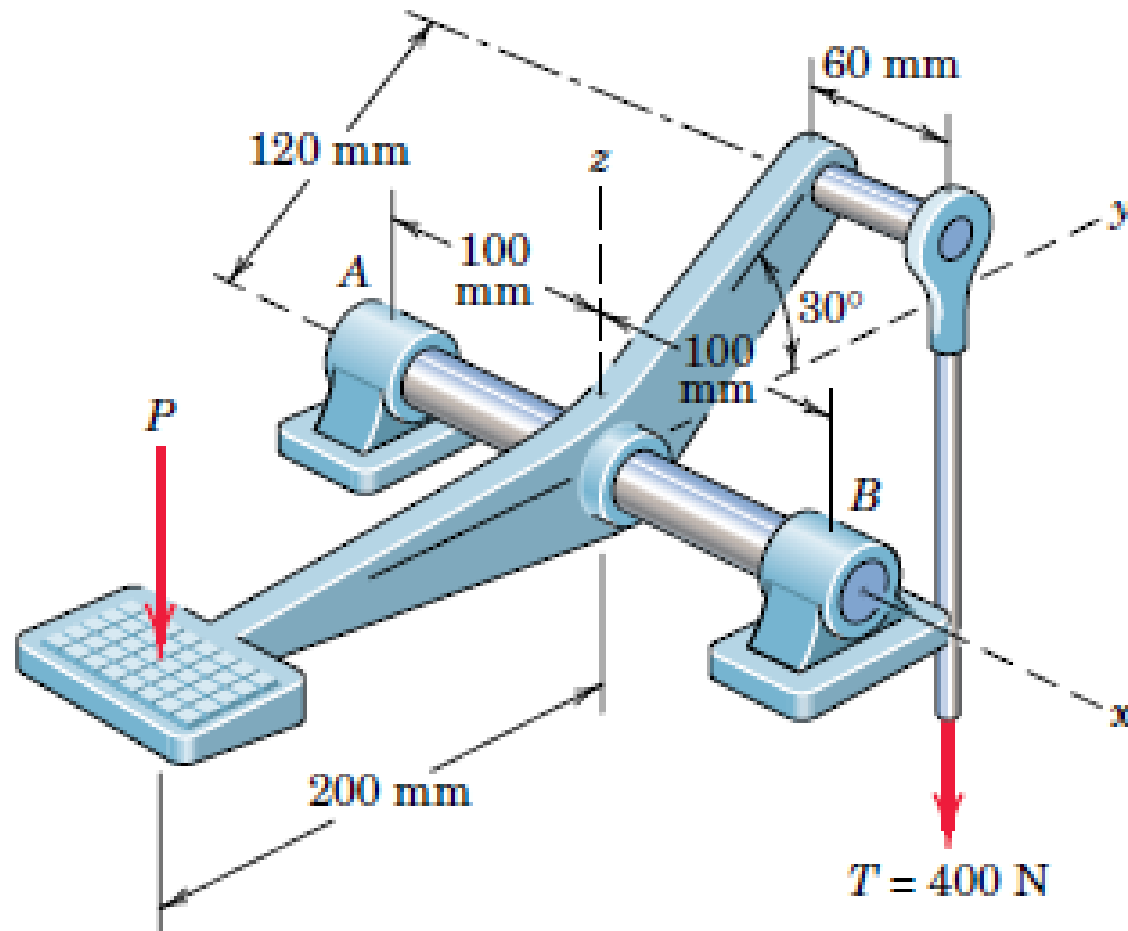


$\Sigma M_B = 0;$
 $200 + 30(2) - 5A_y = 0$
 $A_y = 52 \text{ lb}$

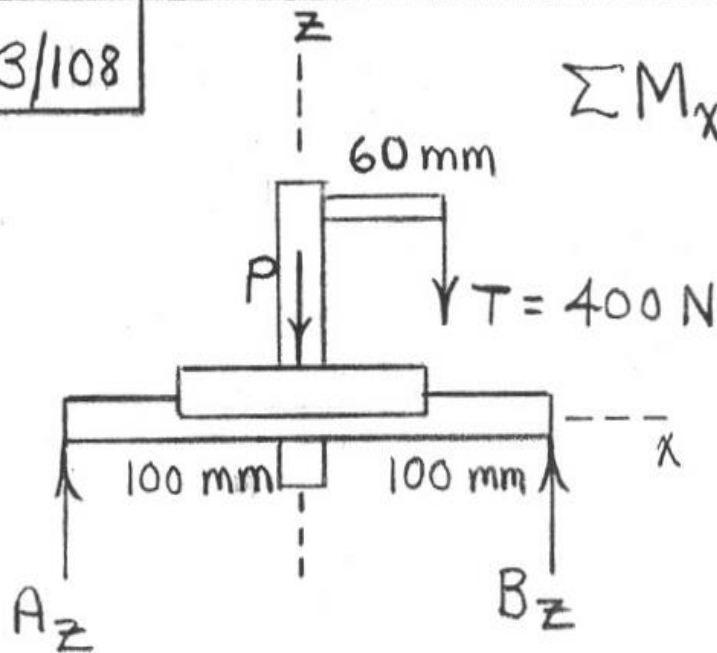
$\Sigma F_y = 0; B_y = 52 \text{ lb}$
 $\Sigma F_x = 0; B_x = 30 \text{ lb}$

$B = \sqrt{(30)^2 + (52)^2} = \underline{60.0 \text{ lb}}$

3/108 A vertical force P on the foot pedal of the bell crank is required to produce a tension T of 400 N in the vertical control rod. Determine the corresponding bearing reactions at A and B .



3/108



$$\sum M_x = 0: P(200) - 400(120 \cos 30^\circ) = 0$$

$$P = 208 \text{ N}$$

$$\sum M_{Ay} = 0: 208(100) + 400(160) - B_z(200) = 0$$

$$B_z = 424 \text{ N}$$

$$\sum F_z = 0: A_z + 424 - 208 - 400 = 0, A_z = 183.9 \text{ N}$$

$$\text{Because } A_y = B_y = 0, \underline{A = A_z = 183.9 \text{ N}}, \underline{B = B_z = 424 \text{ N}}$$



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